The Unequal Weight of Genes: How Early Life Environments Modify the Effects of Genetic Predispositions

Shutian Liu

Latest version

Abstract

Children from poorer families show a stronger link between genetic predictors and body weight. This gradient is widely interpreted as disadvantaged environments amplifying genetic risk. This paper tests an alternative explanation: that conventional genetic predictors inadvertently capture family influences—parental health behaviours, dietary environments, and other conditions correlated with both parental genetics and child weight—so the gradient reflects these family channels rather than children's own genes becoming more consequential under disadvantage. Using linked parent–child genotype data from the UK Millennium Cohort Study, I compare two genetic predictors that differ in their exposure to these family influences: a conventional predictor and one that isolates the child's directly inherited genetic effects. The conventional predictor shows a clear socioeconomic gradient; the direct-effects predictor shows none. Cross-outcome comparisons reveal that the mechanism is trait-specific: family-level pathways drive the gradient for BMI, point estimates for mental health are suggestive of children's own genetic effects playing a role, and neither channel generates a gradient for smoking. The results redirect the policy target from children's individual genetic risk toward the household environments through which the gradient operates.

I. Introduction

Socioeconomic gradients in childhood obesity are large, persistent, and a major public-health concern in most high-income countries (Autret and Bekelman, 2024), with downstream consequences for health, labour market outcomes, and wellbeing (Lindeboom et al., 2010; Segal et al., 2021). An influential body of work using polygenic indices (PGIs)—composite genetic scores that aggregate the BMI-relevant effects of many common DNA variants into a single predictor—has consistently shown that the PGI–BMI association is substantially stronger among children from lower-SES households than among those from higher-SES households (Hüls et al., 2021; Bann et al., 2022). The finding is typically read as disadvantaged environments "unlocking" or "amplifying" children's genetic susceptibility to weight gain, carrying an implicit policy prescription: compress the environmental gradient and the genetic gradient will compress with it.

This interpretation treats as resolved an identification problem that the conventional design cannot solve. A standard PGI is built from population-GWAS weights that bundle three distinct genetic channels at each variant (Kong et al., 2018): the child's own direct effect on BMI, the indirect effects of parental genotypes operating through parenting and household environments (Wertz et al., 2019;

Breinholt and Conley, 2023), and residual population stratification. A steeper PGI–BMI slope at low SES is therefore consistent with at least two fundamentally different causal stories. Under *direct-effect moderation* (H1), disadvantage amplifies the within-child translation of genetic risk into body weight: the same alleles produce larger BMI differences at low SES. Under *family-channel moderation* (H2), the interaction is an artefact of what a standard PGI measures: because a child's PGI partly reflects inherited parental genotypes, it is correlated with parental behaviours, household food environments, and resource allocation; if those family channels are stronger at low SES, a standard PGI will appear to predict BMI more steeply among disadvantaged children even when children's own biology responds identically to environment. The policy implications of the two stories diverge sharply: H1 motivates interventions that target the environments children experience, H2 motivates interventions that target the households they grow up in.

This paper separates the two channels empirically. Using linked child–mother–father genotypes in the UK Millennium Cohort Study, we construct two polygenic indices for BMI in the same children and compare how each moderates the BMI–SES association. The *Standard PGI* uses the population-GWAS weights of Yengo et al. (2018) ($N \approx 700{,}000$), which aggregate all three channels. The *direct genetic effects (DGE) PGI* uses the family-GWAS weights of Tan et al. (2026), whose SNP-level estimates isolate the child's own causal alleles by exploiting within-family Mendelian segregation at the discovery stage. Contrasting the SES interactions of the two predictors, while controlling for maternal and paternal PGIs to absorb residual family-level genetic background, identifies which channel the observed PGI×SES gradient loads on. To our knowledge, this is the first empirical Standard-vs-DGE decomposition of the PGI×SES gradient for childhood BMI.

The two predictors produce sharply different answers. The Standard PGI reproduces the well-documented gradient: the implied PGI–BMI slope is 0.28 BMI units per standard deviation steeper in the lowest income quintile than in the highest, with the interaction coefficient statistically significant and stable across specifications. The DGE PGI shows no such gradient—interaction coefficients are close to zero in every specification and do not generate a stable income profile in the direct genetic association with BMI. Because the two predictors differ only in whether their SNP-level weights reflect all three genetic channels or only the direct-effect channel, the contrast localises the empirical PGI×SES gradient to family-level channels rather than to differential expression of children's own genetic risk. H2 is supported over H1.

Three additional analyses sharpen this conclusion. First, allowing moderation to vary by age places the Standard-PGI gradient in late adolescence, whereas the DGE interaction is flat across developmental stages. Second, a non-parametric specification in income quintiles reveals a monotone slope profile for the Standard PGI—steeper at the bottom, flatter at the top—that is absent for the DGE PGI. Third, applying the same Standard-vs-DGE comparison to two additional outcomes shows that the BMI pattern is not a mechanical artefact of the research design: ever-smoking at age 17 shows no PGI×SES moderation for either predictor, while adolescent mental health difficulties show a directional contrast in the opposite direction to BMI—the DGE predictor carries the larger interaction point estimates, suggestive of H1-type processes operating through direct psychological channels. Restricting the analysis to families with both parents directly genotyped, which removes any contribution from Mendelian-imputed parents, preserves and in several specifications strengthens the main BMI result.

These findings have two implications. First, the widely replicated PGI×SES gradient for childhood BMI is primarily informative about family environments, not about how children's own genetic risk is modified by deprivation. Policies that aim to compress the gradient should therefore target the household—parental health behaviours, food environments, resource allocation—rather than attempt to "buffer" children's biology against environmental adversity. Second, the divergence between Standard and DGE PGIs estimated in the same sample cautions against interpreting PGI×E interactions in conventional population-GWAS designs as evidence of biological gene-by-environment effect modification without a family-based comparison.

The remainder of the paper is organised as follows. Part II reviews the genetic background, the measurement and identification challenges of PGI-based gene–environment interaction analyses, and the related literature. Part III formalises H1 and H2 and derives their observable predictions. Part IV describes the data and empirical strategy. Part V presents results. Part VI discusses the findings and concludes.

II. Background and Related Literature

1 Polygenic Indices: Construction and Causal Interpretation

1.1 The GWAS Statistical Framework

Human genetic variation arises at millions of positions in the genome where individuals carry different nucleotide bases. A single nucleotide polymorphism (SNP) is one such variable position. At each SNP, an individual carries two alleles—one inherited from each parent—and the standard convention is to code the genotype as the count of copies of a designated reference allele, $x_i \in \{0, 1, 2\}$. Genome-wide association studies (GWAS) estimate, one SNP at a time, the association between this count and a phenotype of interest.

For a continuously distributed outcome such as BMI, the association at SNP ℓ is estimated by regressing the outcome on the allele count:

$$y_i = \alpha + \beta_\ell \, x_{i\ell} + \boldsymbol{\gamma}' \mathbf{c}_i + \varepsilon_i, \tag{1}$$

where y_i is BMI, $x_{i\ell} \in \{0, 1, 2\}$ counts copies of the reference allele at SNP ℓ, $\mathbf{c}_i$ collects standard controls (sex, age, genetic principal components that capture broad ancestry differences), and ε_i is the residual. The OLS estimate $\hat{\beta}_\ell$ is the average difference in BMI associated with carrying one additional copy of the reference allele, conditional on $\mathbf{c}_i$. This regression is run separately for every SNP across the genome—typically several million positions—yielding a set of effect estimates $\{\hat{\beta}_\ell\}$.

Because any single SNP explains only a small fraction of phenotypic variation in complex traits like BMI, individual SNP estimates are uninformative predictors. A polygenic index (PGI) addresses this by aggregating across all included SNPs:

$$\widehat{\mathrm{PGI}}_i = \sum_{\ell=1}^{L} \hat{\beta}_\ell \, x_{i\ell}. \tag{2}$$

The PGI is a weighted sum of allele counts, using GWAS effect estimates as weights. By pooling signal from thousands to millions of variants, it produces a composite measure of genetic predisposition toward the trait. A higher BMI PGI indicates that an individual carries more BMI-associated alleles, weighted by their estimated effects. PGIs constructed in this way have become widely used in social science research as measures of genetic endowment, included alongside socioeconomic and demographic characteristics in standard regression analyses (Benjamin et al., 2024; Morris et al., 2024).

2 Identification and Measurement Challenges in Polygenic Index Analyses

Using a PGI as a regressor in gene–environment interaction analysis introduces two distinct classes of challenges that affect different aspects of inference. The first is a measurement challenge: because PGI weights are estimated from finite GWAS samples, the resulting score is a noisy proxy for the latent additive genetic component it targets. The second is an identification challenge: even a perfectly measured PGI conflates the individual's direct genetic effects with indirect parental channels and population stratification confounding, so the composite score aggregates multiple causal pathways into a single predictor (Kong et al., 2018). The distinction matters because the two challenges require fundamentally different solutions. Measurement imprecision can be mitigated through better construction methods and statistical corrections applied at the estimation stage. The identification problem cannot—it requires a different source of genetic variation at the GWAS design stage.

The measurement challenge takes the form of classical errors-in-variables bias. Becker et al. (2021) formalise the PGI as a rescaled noisy proxy for the latent additive SNP factor, where the degree of measurement error is governed by the ratio of the PGI's predictive accuracy (R^2) to the underlying SNP heritability (h^2_{SNP}). In univariate regression, this attenuates the PGI coefficient toward zero; in multivariate settings, the bias extends to coefficients on covariates and can operate in either direction. For gene–environment interaction studies, the implication is particularly consequential: measurement error in the PGI propagates into the interaction term, attenuating estimated gene–environment interactions and potentially understating the magnitude of true effect heterogeneity across socioeconomic strata (Wang et al., 2020). Two families of correction are available. On the construction side, Bayesian shrinkage methods such as LDpred2 (Privé et al., 2021) model linkage disequilibrium structure explicitly and shrink effect estimates toward their posterior means, reducing winner's curse inflation (Shi et al., 2016) and improving out-of-sample prediction. On the estimation side, measurement error can be corrected using instrumental variable strategies: ORIV constructs two PGIs from independent halves of the discovery GWAS and uses each as an instrument for the other (Muslimova et al., 2024); PGI-RC applies repository-based correction factors calibrated to each phenotype (Becker et al., 2021; Alemu et al., 2025).

A related concern is that PGI predictive accuracy varies across populations and, more subtly, across socioeconomic strata within a single ancestry group. PGIs trained in European-ancestry samples lose predictive power when applied to genetically distant populations (Martin et al., 2019; Wang et al., 2020); this portability problem is less relevant in the present study, where both the discovery GWAS and the MCS sample are of European ancestry. However, Mostafavi et al. (2020) show that even within an ancestry group, prediction accuracy varies across demographic and socioeconomic strata, tracking differences in SNP heritability. If the PGI is a systematically noisier predictor at lower SES, differential measurement error could generate or inflate a PGI $\times$ SES interaction in the absence of true effect modification—a concern directly relevant to this paper's empirical question.

2.1 What the GWAS Coefficient Actually Captures

The identification challenge is more fundamental than measurement error, and cannot be resolved by better estimation techniques. A central objective of standard GWAS is to estimate the direct genetic effect of a variant—conceptually, a variant substitution effect: the counterfactual change in an individual's phenotype if their genotype at a given locus were altered from conception, holding everything else constant.[1] In practice, GWAS approximates this by testing whether allele-frequency differences at each SNP are associated with phenotype differences across individuals. A causal interpretation requires that these differences arise because the SNP (or a nearby variant in linkage disequilibrium) truly influences the trait. However, many other forces can generate allele-frequency differences across individuals, thereby violating this assumption.

To formalise the problem, suppose the true data-generating process at SNP ℓ is

$$y_i = \beta_\ell^{\text{dir}}\, x_{i\ell} + \eta_\ell\, Z_i + u_i, \tag{3}$$

where $x_{i\ell} = G_{i\ell}$ is the child's allele count, β_ℓ^{dir} is the direct causal effect of the child's own genotype, and Z_i captures any factor that (i) influences the trait ($\eta_\ell \neq 0$) and (ii) creates systematic differences in genotype distributions across individuals ($\text{Cov}(x_{i\ell}, Z_i) \neq 0$). If Z_i is excluded from the regression, the standard omitted variable formula gives

$$\hat{\beta}_\ell \xrightarrow{p} \beta_\ell^* = \beta_\ell^{\text{dir}} + \eta_\ell \cdot \frac{\text{Cov}(x_{i\ell},\, Z_i)}{\text{Var}(x_{i\ell})}. \tag{4}$$

The bias term is non-zero whenever both conditions hold simultaneously. The key empirical question is therefore: what kinds of real-world forces could both shape BMI and make some groups of people more likely to carry certain genotypes? Two broad classes of variable satisfy both conditions.

Indirect genetic effects (genetic nurture). The first class is parental genotypes. Because children inherit alleles from their parents, a child's allele count at any SNP is mechanically correlated with parental allele counts, satisfying condition (ii). Parental genotypes independently influence child BMI—not through the alleles they transmit, but through the household environment they create—satisfying condition (i).

To fix ideas, consider two children who carry the identical genotype at a BMI-associated SNP. If one child's parents carry genotypes associated with health-conscious behaviour and structured dietary routines, while the other child's parents do not, the first child will tend to have lower BMI—even though the children's own alleles are the same.

This intuition can be formalised in three steps. First, parental genotypes shape the family environment:

$$E_i = \alpha_p\, G_{pi\ell} + \alpha_m\, G_{mi\ell} + U_i^E, \tag{5}$$

where $G_{pi\ell}$ and $G_{mi\ell}$ are the father's and mother's allele counts at SNP ℓ, and E_i captures environmental inputs—diet quality, physical activity norms, household routines—that parental traits

[1] This interpretation targets the biological pathway from the child's own alleles to the phenotype. It excludes any influence that operates through parental genotypes, household environments, or other family and environmental channels.

linked to genotype (education, health behaviours) generate. Second, the child's BMI depends on both their own direct genetic effect and this environment:

$$y_i = \beta_\ell^{\text{dir}}\, G_{i\ell} + \gamma\, E_i + U_i^Y. \tag{6}$$

Third, substituting (5) into (6) yields

$$y_i = \beta_\ell^{\text{dir}}\, G_{i\ell} + \underbrace{\gamma\alpha_p}_{=\eta_p}\, G_{pi\ell} + \underbrace{\gamma\alpha_m}_{=\eta_m}\, G_{mi\ell} + \tilde{U}_i, \tag{7}$$

so that parental genotypes predict child BMI even when not all parental alleles are transmitted. This parent $\to$ environment $\to$ child pathway is the indirect genetic effect, or genetic nurture (Kong et al., 2018). Because roughly half of parental alleles are transmitted, $\text{Cov}(G_{i\ell}, G_{pi\ell}) \neq 0$ and $\text{Cov}(G_{i\ell}, G_{mi\ell}) \neq 0$, so part of the observed SNP–BMI association in a standard GWAS reflects these parental environmental influences rather than the child's own biology.

Population stratification. The second class is an individual's ancestral background, which satisfies both conditions through a sequential mechanism. First, non-random mating—including both assortative mating on traits correlated with BMI and geographic clustering of individuals with shared ancestry—concentrates similar-ancestry individuals within social and geographic clusters, causing the reference allele frequency at every SNP to vary across clusters. Formally, let Z_i index individual i's ancestral cluster; then the expected allele count at SNP ℓ is $\mathbb{E}[x_{i\ell} \mid Z_i] = 2p_\ell(Z_i)$, where $p_\ell(Z_i)$ is the reference allele frequency in cluster Z_i. Because p_ℓ varies across clusters, $\text{Cov}(x_{i\ell}, Z_i) \neq 0$ for essentially every SNP in the genome, satisfying condition (ii). Second, the same ancestry clusters differ systematically in social and physical environments—dietary traditions, income levels, neighbourhood characteristics—that causally affect BMI, so $\text{Cov}(y_i, Z_i) \neq 0$, satisfying condition (i). The genome-wide pattern of between-cluster allele frequency differences therefore induces spurious SNP–BMI associations that reflect environmental heterogeneity across ancestral groups rather than any SNP's own causal effect. Standard GWAS includes genetic principal components in $\mathbf{c}_i$ to absorb stratification, but residual confounding may persist when principal components do not fully capture fine-grained ancestry differences.

Three-channel decomposition. Applying (4) to each class and collecting terms, the GWAS coefficient in a population-based sample converges to

$$\beta_\ell^* = \underbrace{\beta_\ell^{\text{dir}}}_{\text{direct effect}} + \underbrace{\beta_\ell^{\text{ind}}}_{\text{indirect (nurture)}} + \underbrace{\beta_\ell^{\text{conf}}}_{\text{stratification}}, \tag{8}$$

and since the PGI in (2) is linear in these weights, it inherits the same additive structure:

$$\widehat{\text{PGI}}_i = \underbrace{\widehat{\text{PGI}}_i^{\text{dir}}}_{\text{direct}} + \underbrace{\widehat{\text{PGI}}_i^{\text{ind}}}_{\text{indirect}} + \underbrace{\widehat{\text{PGI}}_i^{\text{conf}}}_{\text{stratification}}. \tag{9}$$

Alemu et al. (2025) formalise this conflation as a non-classical source of bias that persists even in the limit of infinite GWAS samples: the population-based PGI does not converge to the direct-effect PGI because the underlying weights capture associational rather than causal effects. The severity varies by phenotype: within-sibship GWAS estimates for educational attainment are approximately

47 percent smaller than population estimates, while BMI shows limited shrinkage (Howe et al., 2022). Yet even a modest indirect component can be consequential for gene–environment interaction analysis if that component is specifically correlated with socioeconomic environments—precisely the scenario in which a standard PGI × SES interaction would reflect family-level channels rather than moderation of direct genetic effects. Table 1 summarises the three channels, their sources, and their relationship to the family-based estimation strategy used in this paper.

Table 1: Components of the standard GWAS effect estimate

Channel	**Source**	**Example for BMI**	**Isolated by family GWAS?**
Direct effect (β^{dir})	Child's own alleles acting through biological pathways	FTO rs9939609 reduces satiety signalling in the hypothalamus; each additional risk allele increases energy intake and raises BMI by roughly $0.4\,\text{kg/m}^2$ through the child's own physiology	Yes
Indirect effect (β^{ind})	Parental alleles influencing the home environment (genetic nurture)	A parent carrying FTO risk alleles tends to purchase more energy-dense foods and keep larger portion sizes; children in such households accumulate higher BMI through shared diet, regardless of which FTO alleles they personally inherited	No—absorbed by parental genotype controls
Stratification (β^{conf})	Non-random mating clusters similar-ancestry individuals together; those clusters occupy distinct social environments that independently affect BMI	South Asian and Northern European populations differ in reference allele frequencies across thousands of SNPs and in dietary patterns that causally affect BMI. A mixed-ancestry GWAS detects a SNP–BMI association that is partly spurious: the alleles predict BMI because they tag the ancestry group, which predicts the diet	No—absorbed by parental genotype controls

Notes: Standard GWAS in samples of unrelated individuals estimates the sum of all three components (equation 8). Family-based GWAS, which conditions on parental genotypes, isolates the direct genetic effect by exploiting within-family variation in offspring genotypes generated by random Mendelian segregation. The DGE PGI, constructed from family-based GWAS estimates that isolate direct genetic effects, allows the first channel to be evaluated separately from the other two.

This paper addresses both categories of challenge. On the measurement side, the standard BMI PGI is constructed using LDpred (Vilhjálmsson et al., 2015), which mitigates construction-related attenuation through Bayesian shrinkage of GWAS weights. The final specification will further incorporate ORIV correction to address residual attenuation in the interaction estimates. On the identification side, the paper uses a DGE PGI constructed from family-based GWAS weights (Howe et al., 2022; Tan et al., 2026), isolating variation attributable to children's own directly inherited alleles. The following subsection describes this family-based identification strategy in detail.

2.2 Family-Based Identification of Direct Genetic Effects and Parental Imputation

Family-based GWAS methods isolate β^{dir} by conditioning on parental genotypes, exploiting a fundamental property of Mendelian inheritance: conditional on both parents' genotypes, the alleles a child receives are determined by random segregation in meiosis. Each parent independently transmits one of their two alleles to the child with equal probability, regardless of the family environment. This random draw generates within-family variation in offspring genotypes that is orthogonal to all environmental factors correlated with parental alleles.

Formally, including parental allele counts $g_{pi\ell}$ and $g_{mi\ell}$ as controls in the GWAS regression

$$y_i = \delta_\ell \, g_{ci\ell} + \alpha_{p\ell} \, g_{pi\ell} + \alpha_{m\ell} \, g_{mi\ell} + \boldsymbol{\gamma}' \mathbf{c}_i + \varepsilon_i \tag{10}$$

absorbs the between-family genetic variation that co-moves with parental environments. The residual variation in $g_{ci\ell}$—the segregation residual $u_{i\ell} = g_{ci\ell} - \frac{1}{2}(g_{pi\ell} + g_{mi\ell})$—is generated solely by the random transmission of parental alleles in meiosis and satisfies $\mathbb{E}[u_{i\ell} \mid g_{pi\ell}, g_{mi\ell}] = 0$. Under standard conditions, OLS estimation of (10) consistently identifies the direct genetic effect δ_ℓ (Young et al., 2022). A polygenic index constructed by applying these family-based weights to the child's allele counts—the direct genetic effects PGI (DGE PGI)—therefore predicts BMI through the child's own causal alleles, with the indirect and confounding components absorbed by the parental controls.

The comparison between the standard PGI and the DGE PGI across socioeconomic strata is the central identification strategy in this paper. Because the standard PGI reflects all three channels in equation (8) while the DGE PGI isolates β^{dir}, the two predictors deliver different estimates of $\hat{\beta}_{GS}$ in (12) only if indirect or confounding channels contribute to the observed heterogeneity. A substantial attenuation of $\hat{\beta}_{GS}$ when moving from the standard PGI to the DGE PGI therefore indicates that socioeconomic heterogeneity in the standard predictor is partly driven by family-level channels rather than by environmental modification of children's direct genetic risk.

Both constructing the DGE PGI and conditioning on parental genotypes in the regression require observed or imputed parental genetic data. Implementing this strategy therefore depends on the availability of family genotype information. In the MCS, maternal genotypes are broadly available, but a non-trivial share of fathers were not genotyped. For these families, paternal genotypes can be recovered using a procedure based on Mendelian transmission rules and population allele frequencies (Young et al., 2022). At each SNP ℓ, the imputed paternal genotype is

$$\hat{g}_{pi\ell} = \mathbb{E}\big[\, g_{pi\ell} \mid g_{ci\ell}, \, g_{mi\ell} \,\big], \tag{11}$$

computed under Mendelian transmission and population-level Hardy–Weinberg allele frequencies. Summing over SNPs yields an imputed paternal PGI equal to the conditional expectation of the true paternal PGI given available family information. This imputed quantity is unbiased for the true parental PGI, uses only pre-determined genetic and pedigree information, and is independent of the outcome and socioeconomic covariates; it therefore does not alter the identification logic of the empirical design. The main consequence of imputation is reduced precision—standard errors for paternal coefficients are larger than they would be with fully observed paternal genotypes—but the estimands are unaffected. Robustness checks confirm this: restricting the sample to families with fully observed child–mother–father trios yields larger, not smaller, estimates of socioeconomic heterogeneity in the standard PGI slope, consistent with imputation introducing attenuation rather than generating spurious interactions. Figure 1 summarises both identification designs side by side.

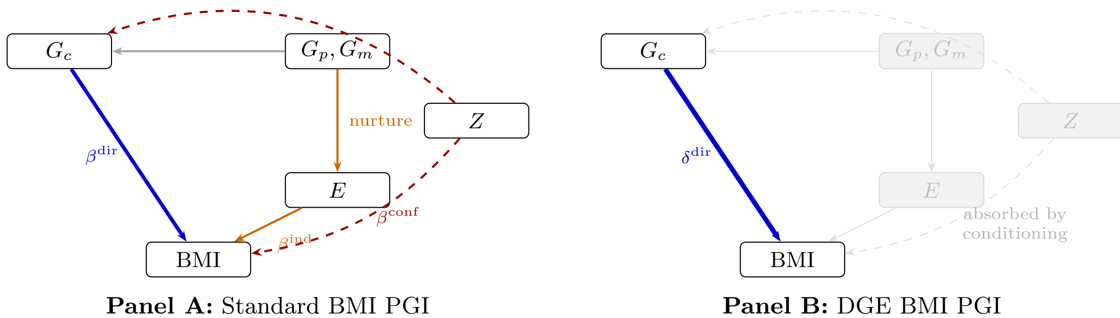


Figure 1: Channels captured by the standard BMI PGI versus the direct-genetic-effects (DGE) BMI PGI.

Notes: **Panel A** depicts the three pathways bundled into a standard GWAS-based PGI: (i) the child's direct genetic effect on BMI (β^{dir}, blue); (ii) indirect effects transmitted through the family environment associated with parental genotypes G_p, G_m (β^{ind}, orange); and (iii) stratification confounding from co-variation of genetic ancestry with social environment (β^{conf}, red dashed). **Panel B** depicts the DGE PGI, constructed from family-based GWAS that conditions on parental genotypes. Conditioning absorbs the indirect and stratification channels (faded grey), leaving only the direct genetic effect (δ^{dir}, blue).

With the identification strategy established, the following section reviews the empirical evidence on gene–environment interactions and explains why existing studies cannot resolve the ambiguity between H1 and H2.

3 Gene × Socioeconomic Environment Interactions in Health Outcomes

3.1 Interpreting a PGI × SES Interaction

The decomposition in equations (8)–(9) has direct statistical consequences for any regression that interacts the PGI with a socioeconomic variable. Consider the canonical gene–environment model used in this paper:

$$\text{BMI}_{it} = \alpha + \beta_G \widehat{\text{PGI}}_i + \beta_S \,\text{SES}_i + \beta_{GS} \left(\widehat{\text{PGI}}_i \times \text{SES}_i\right) + \delta_t + \mathbf{X}_i' \boldsymbol{\gamma} + \varepsilon_{it}. \tag{12}$$

Because $\widehat{\text{PGI}}_i$ is a composite of three components (equation 9), the interaction term is also composite:

$$\widehat{\text{PGI}}_i \times \text{SES}_i = \widehat{\text{PGI}}_i^{\text{dir}} \times \text{SES}_i + \widehat{\text{PGI}}_i^{\text{ind}} \times \text{SES}_i + \widehat{\text{PGI}}_i^{\text{conf}} \times \text{SES}_i. \tag{13}$$

The estimated coefficient $\hat{\beta}_{GS}$ therefore captures a weighted combination of how each channel's association with BMI varies across the socioeconomic distribution. A non-zero $\hat{\beta}_{GS}$ does not imply that the child's direct genetic effects are modified by SES: it is equally consistent with SES moderating indirect family-level channels, or with residual stratification confounding that co-varies with socioeconomic position. The three explanations carry fundamentally different interpretations and imply different policy conclusions. Table 2 formalises this.

Table 2: Interpreting a PGI × SES interaction under alternative channels

Channel driving $\hat{\beta}_{GS}$	**Substantive interpretation**	**Policy implication**
Direct genetic effects	Socioeconomic disadvantage amplifies (or buffers) the expression of children's own genetic predispositions toward higher BMI	Interventions should target environmental conditions that interact with individual-level biological risk
Indirect genetic effects (nurture)	Disadvantaged households translate parental genetic predisposition toward obesity more strongly into the home environment, independently of the child's own alleles	Interventions should target household resources and parental behaviours rather than child-level genetic risk
Stratification confounding	Genetic ancestry and socioeconomic position co-vary in a way not fully absorbed by principal components, producing a spurious interaction	Observed heterogeneity may reflect measurement artefacts rather than genuine gene–environment moderation

Notes: A non-zero $\hat{\beta}_{GS}$ from a standard PGI (equation 12) is consistent with all three channels simultaneously, since the PGI bundles their contributions. The DGE PGI, constructed from family-based GWAS estimates that isolate direct genetic effects, allows the first channel to be evaluated separately from the other two.

A large empirical literature in economics, public health, and medicine tests whether polygenic predictors for BMI and related health traits exhibit heterogeneous associations across socioeconomic status and other upstream social environments. Biroli et al. (2026) provide a systematic framework for classifying these studies along two dimensions: the exogeneity of the environmental variable and the exogeneity of the genetic predictor, emphasising that credible causal inference about gene–environment interplay requires attention to both. A typical empirical specification regresses a health outcome on a polygenic score, an environment measure such as education or disadvantage, and their interaction, often with demographic controls and genetic principal components (Carter et al., 2022; Pereira et al., 2022). The motivating idea is that environments may alter the mapping from genetic predisposition to realised outcomes (Hüls et al., 2021). In practice, the interpretation of the interaction depends on whether the environment can be treated as plausibly exogenous and on whether the polygenic predictor can be interpreted as capturing a stable genetic component that is comparable across social strata (Biroli et al., 2026). A first organising distinction is between studies where the environment is induced by a plausibly exogenous change and studies where the environment is a socioeconomic or neighbourhood measure that is not randomly assigned.

In quasi-experimental designs, the interaction is framed as treatment effect heterogeneity: the effect of a policy or shock differs by genotype (Barcellos et al., 2018). Early contributions exploited compulsory schooling reforms (Barcellos et al., 2018), the minimum legal drinking age (Fletcher and Lu, 2021), and discontinuities around major shocks (Furuya et al., 2022). A common finding is that the presence and direction of gene–environment interactions depends heavily on the specific environmental variation exploited, with some designs detecting significant heterogeneity and others finding none. More recent work extends this approach to health and prenatal exposures, with similarly heterogeneous results: Baker et al. (2024) find that advantageous early-life environments cushion genetic risk for heart disease (interaction survives within-family analysis); van den Berg et al. (2025) find only modest gene–environment interactions for BMI in a natural experiment on sugar rationing; and Dias Pereira et al. (2025) find no interaction between maternal smoking and the child's birth weight PGI using Mendelian randomisation. Muslimova et al. (2024) find gene–investment complementarity for educational attainment using within-family variation and ORIV-corrected estimates.

Across this set of studies, two patterns emerge. First, a genotype–environment interaction in a quasi-experimental setting is interpreted as heterogeneity in the causal impact of an external change, which is conceptually distinct from claiming that SES directly "activates" genetic risk (Barcellos et al., 2018). Second, even with plausibly exogenous environmental variation, results vary considerably across traits, populations, and designs—suggesting that gene–environment interaction is not a universal phenomenon but depends on the specific biological and social pathways involved.

In observational cohort studies using education, income, deprivation, or neighbourhood exposures as the environment, evidence on BMI PGI interactions is mixed in both direction and statistical detectability (Bann et al., 2022). Several papers report patterns consistent with amplification under disadvantage: the PGI–BMI slope is larger in lower parental education groups or in higher disadvantage settings, and in some settings the interaction is concentrated among individuals at higher polygenic risk (Hüls et al., 2021). Other papers find additive patterns in which the PGI gradient and the socioeconomic gradient are both present, but the interaction term is near zero across the life course, across overweight trajectories, or across cohorts (de Roo et al., 2023). Differences in results often coincide with differences in the age range studied, the cohort and institutional context, and the construction of the phenotype (continuous BMI versus overweight classes or trajectories). Beyond BMI, the pattern varies across outcomes and contexts. For educational attainment, Houmark et al. (2025) document that genetic and socioeconomic achievement gaps are similar in magnitude in Danish elementary schools, and that the gene–SES interaction shifts direction across the developmental trajectory. Isungset et al. (2022) find no or very small gene–SES interactions for educational performance in a Norwegian welfare state using a trio design, suggesting that strong social institutions may mute the interplay. Uchikoshi and Conley (2021) find weak evidence that parental SES amplifies the education PGI's effect on course tracking in U.S. secondary schools. For smoking, Bierut et al. (2023) find that gene–environment interactions depend critically on how the environment is measured: childhood financial distress moderates the smoking PGI, but parental education does not.

Two technical features recur across this literature and affect what the interaction coefficient represents. First, conclusions about effect modification can be scale dependent . For BMI, which is continuous, the interaction is often interpreted on the additive scale (differences in mean BMI per unit of PGI) (Bann et al., 2022). For obesity as a binary outcome, studies may report interactions on multiplicative scales (odds ratios or relative risks), and the presence or absence of interaction can change across scales even when underlying data and covariates are unchanged (Carter et al., 2022). Several studies therefore emphasise that "interaction" is not a single object, but depends on the outcome scale, the functional form, and the contrast being interpreted.

Second, interpretability hinges on the extent to which gene–environment correlation and residual confounding are addressed (Mason et al., 2020). Socioeconomic status is socially patterned and correlated with geographic location, family background, and behavioural environments (Mason et al., 2020). These factors can correlate with genetic structure through ancestry, migration, assortative mating, and social stratification. As a result, an observed PGI × SES interaction can arise because the environment is endogenous to genotype, because the genetic predictor captures family correlated channels, or because the observed environment measure proxies unobserved household conditions that co-vary with genotype (Carter et al., 2022). Many studies mitigate these concerns using genetic principal components, rich covariate adjustment, restricting to within-ancestry samples, using pre-determined childhood SES, and sensitivity analyses (Bann et al., 2022). Family fixed effects designs in twins or siblings are sometimes used to reduce shared family confounding (Amin et al., 2018). Nonetheless, papers commonly note that residual confounding remains plausible when SES and neighbourhood environments are measured observationally and when genetic predictors are derived from GWAS in unrelated samples (Mason et al., 2020).

A third recurring issue is what the interaction is "about". Some papers interpret a positive interaction between genetic risk and an obesogenic environment as consistent with a diathesis–stress narrative, where disadvantage increases the expression of genetic predisposition (Hüls et al., 2021). Others interpret interactions as heterogeneity in behavioural responses or policy responsiveness, rather than biological amplification (Fletcher and Lu, 2021). For example, in settings where education is shifted by a reform, a genotype interaction is often interpreted as heterogeneity in the reform's effect on BMI or health behaviours among compliers, which is conceptually different from claiming that SES itself modifies genetic effects (Barcellos et al., 2018). Similarly, in neighbourhood studies using fast-food proximity or other built environment proxies, the interaction may reflect heterogeneous exposure, heterogeneous behavioural substitution, or heterogeneous measurement error correlated with socioeconomic position (Mason et al., 2020).

Across studies, the strongest interpretability claims typically arise when the environment is plausibly exogenous and when the estimand is explicitly framed as a causal effect of the environment that varies by genotype (Barcellos et al., 2018). When SES is used as the environment, the interaction is more often treated as descriptive heterogeneity that is consistent with multiple mechanisms (Bann et al., 2022). A common implication in this literature is therefore that documenting a PGI × SES interaction is not sufficient to determine whether it reflects modification of direct genetic effects or household and family correlated pathways embedded in the polygenic predictor (Amin et al., 2018).

The gap this paper addresses is the ambiguity that runs through the observational cohort literature: a non-zero PGI × SES interaction is consistent with direct-effect moderation, family-channel moderation, or residual stratification confounding, and existing studies cannot separate these explanations because they rely on standard population-based PGIs that bundle all three channels (Biroli et al., 2026; Benjamin et al., 2024). Several recent papers identify this limitation explicitly: van den Berg et al. (2025) note that their gene–environment estimates "may be capturing the additional exposure driven by the parental genetic predisposition"; Uchikoshi and Conley (2021) call for future studies to distinguish direct genetic mechanisms from genetic nurture in PGI interaction estimates; and Pereira et al. (2022) characterise family-based gene–environment analysis as the field's "ideal experiment" that has yet to be implemented for health outcomes. This paper provides that implementation. By comparing socioeconomic heterogeneity across two predictors that differ precisely in their exposure to the indirect and confounding channels—a standard child BMI PGI and a direct genetic effects PGI constructed from family-based GWAS estimates that condition on parental genotypes—it supplies the identification step that existing PGI × SES studies lack. The Conceptual Framework section below formalises what each hypothesis predicts for this comparison.

III. Conceptual Framework

The literature review established that existing PGI × SES studies cannot discriminate between the mechanisms generating a non-zero interaction. We narrow the problem to two hypotheses that are both consistent with the observed BMI gradient but make distinct predictions for the DGE PGI comparison developed in Section 1.

H1: Direct-effect moderation. Socioeconomic disadvantage amplifies the expression of children's own genetic predispositions toward higher BMI. The biological pathways through which inherited alleles affect body weight—appetite regulation, metabolic efficiency, hormonal signalling—are less effectively offset under resource-constrained conditions: poorer diet quality, chronic stress, disrupted sleep, and fewer opportunities for physical activity each allow genetic differences to translate more completely into body weight. Under H1, moderation is a property of children's direct genetic effects, so it should persist when the genetic predictor is restricted to Mendelian-segregated variation. The DGE PGI × SES coefficient should be negative and of similar magnitude to the standard PGI interaction.

H2: Family-channel moderation. Socioeconomic moderation reflects parental and household-level pathways embedded in the composite standard PGI rather than amplification of the child's own alleles. Parents who carry BMI-increasing alleles tend to create home environments—through dietary practices, activity routines, and household organisation—that raise child BMI independently of which alleles the child inherited (Kong et al., 2018; Meddens et al., 2021). If these parental influences are stronger, or less buffered, under disadvantage—and if parental investment responses to child genotype are themselves socioeconomically stratified (Breinholt and Conley, 2023)—the standard PGI will appear more strongly moderated by SES through parental channels. Conditioning on parental genotypes in the DGE design absorbs these family-level pathways, so the DGE PGI × SES coefficient should attenuate sharply toward zero relative to the standard PGI interaction.

Table 3 formalises the predictions. Both hypotheses generate the same negative standard PGI × SES coefficient; only the DGE PGI comparison discriminates between them. Two additional analyses extend the test. Under H1, moderation should appear across developmental stages where biological amplification operates, and should generalise to other outcomes with direct-effect channels; under H2, moderation should be concentrated in the composite predictor and attenuate across outcomes once the family channel is removed. Part V reports how the evidence maps onto these predictions.

Table 3: Observable predictions under each hypothesis

Hypothesis	**Standard PGI × SES**	**DGE PGI × SES**
H1: Direct-effect moderation	Negative: disadvantage amplifies genetic predisposition toward higher BMI	Also negative: the same biological mechanism operates through the child's own alleles, which the DGE PGI isolates
H2: Family-channel moderation	Negative: parental genetic channels and household pathways are more consequential under disadvantage	Near zero: conditioning on parental genotypes absorbs the family-level channels driving the standard PGI interaction

Notes: The two hypotheses generate identical predictions for the standard PGI × SES interaction; only the DGE PGI comparison discriminates between them. The DGE PGI isolates variation attributable to random Mendelian segregation by conditioning on parental genotypes (Section 1).

IV. Empirical Strategy

4 Data

4.1 Study population

The Millennium Cohort Study (MCS) is a nationally representative birth cohort of approximately 19,000 children born in the United Kingdom in 2000–2002. Participants are surveyed at ages 5, 7, 11, 14, and 17, providing repeated measures of anthropometric, socioeconomic, and behavioural outcomes across childhood and adolescence. A subset of cohort members and their parents were genotyped, yielding linked child–parent genotype data.

The estimation sample consists of children with non-missing BMI, genotype data, parental genotype data (observed or imputed), baseline socioeconomic information, and control variables. All specifications within each table are estimated on a common sample to ensure comparability across columns.

4.2 Body mass index

The outcome variable is body mass index (BMI), defined as weight in kilograms divided by height in metres squared. Each child contributes up to five BMI observations across the survey waves, yielding a child–wave panel. The data are structured in long format, so the unit of observation in all regressions is a child–wave pair (i, t).

4.3 Polygenic indices

Genetic predisposition toward higher BMI is captured using two child-level polygenic indices that differ in the SNP-level weights used to aggregate them. We construct both indices from publicly available summary statistics applied to the MCS genotype data; the full pipeline is documented in Appendix 7.5.

Standard BMI PGI. SNP weights are drawn from the BMI meta-analysis of Yengo et al. (2018), which combines the GIANT consortium and UK Biobank into a discovery sample of approximately 700,000 individuals of European ancestry. Raw marginal GWAS associations are not suitable as PGI weights because linkage disequilibrium induces correlated effects across neighbouring SNPs, inflating score variance and degrading out-of-sample prediction. We therefore re-weight the summary statistics using LDpred (Vilhjálmsson et al., 2015), a Bayesian method that shrinks each marginal estimate toward zero under an explicit prior on the causal effect-size distribution. We set the prior fraction of causal variants to $p = 1$ (the infinitesimal model), reflecting BMI's highly polygenic architecture. In our MCS estimation sample, the resulting Standard BMI PGI explains 4.9% of cross-sectional BMI variance in a pooled bivariate specification ($\hat{\beta}_G = 0.92$ kg/m^2 per SD; $p < 10^{-140}$), in line with published benchmarks for BMI PGIs built from the same discovery GWAS.

Direct genetic effects (DGE) BMI PGI. SNP weights are drawn from the family-based BMI GWAS of Tan et al. (2026), which estimates the direct genetic effects δ_ℓ of equation 10 by exploiting random Mendelian segregation within families. For BMI specifically, Tan et al. report a direct-effect SNP heritability of $h^2_{\text{direct}} = 0.21$ (SE 0.01) estimated on a median effective discovery sample of approximately 82,000 HapMap3 individuals, comparable to the population-effect heritability they estimate for the same phenotype ($h^2_{\text{pop}} = 0.22$, SE 0.01). The PGI is produced by snipar (Young et al., 2022), software designed for family-based polygenic scoring, which jointly constructs child, maternal, and paternal PGIs from pedigree-linked genotypes and substitutes Mendelian-imputed allele counts for parents whose genotypes are missing (see Section 1, equation 11). In the same MCS estimation sample, the DGE BMI PGI explains only 0.04% of cross-sectional BMI variance ($\hat{\beta}_G = 0.09$ kg/m^2 per SD; $p = 0.02$), roughly two orders of magnitude below the Standard PGI. Appendix 7.5 documents the variant-level processing steps.

Relationship between the two indices. Despite both indices targeting BMI, and despite Tan et al. reporting a genome-wide SNP-level correlation of 0.92 between the direct-effect and population-effect weights for BMI, the Standard and DGE child PGIs are nearly uncorrelated in our MCS sample ($r \approx 0.06$). The gap between the weight-level correlation (≈ 0.92) and the score-level correlation (≈ 0.06) reflects how snipar constructs the DGE score for children in a family-based design: the within-family segregation signal that identifies δ_ℓ is much smaller than the total genetic variation exploited by a population-based PGI, so even when the underlying weight vectors are similar, the resulting child-level scores carry substantially different information. The two indices are therefore not related by a simple rescaling or residualisation, and the DGE PGI must be constructed independently to isolate the direct-effect channel used in our identification strategy.

All child polygenic indices are standardised to mean zero and unit standard deviation in the estimation sample. Maternal and paternal BMI PGIs are produced jointly with the child index by the same snipar run and enter extended specifications as controls for family-level genetic channels.

4.4 Socioeconomic status

Socioeconomic status is measured using equivalised household income at baseline. A household with two adults and three children has higher total costs than a single-adult household earning the same amount, so raw income overstates the larger household's per-person living standard. The modified OECD equivalence scale addresses this by dividing total household income by a household-size adjustment factor: a weight of 1.0 for the first adult, 0.5 for each additional member aged 14 or older, and 0.3 for each child younger than 14. Letting Y_i denote total household income and E_i the sum of these weights, equivalised household income is

$$Y_i^{eq} = \frac{Y_i}{E_i}. \tag{14}$$

For example, a household with two adults and two children under 14 has $E_i = 1.0+0.5+0.3+0.3 = 2.1$, so its equivalised income is total income divided by 2.1. This adjustment places households of different sizes on a comparable scale of material living standards.

Primary specification: income quintiles. The main SES measure groups equivalised household income into quintiles, coded from 1 (lowest) to 5 (highest) and centred at the middle quintile so that $SES_c = SES - 3$. Under this parameterisation, the main effect of the polygenic index is interpreted as the association at the midpoint of the income distribution, while the interaction term captures deviations in the genetic association as income moves away from the middle quintile. Income quintiles are the primary specification because they impose minimal functional-form assumptions on the SES gradient and provide a natural interpretation as differences between discrete socioeconomic groups.

Robustness specification: standardised log equivalised income. As a robustness check, we replace income quintiles with standardised log equivalised income, which preserves continuous variation in household resources and captures proportional rather than rank-based differences. Comparing results across both specifications allows us to assess whether the estimated gene–environment interaction is sensitive to the functional form imposed on socioeconomic status.

4.5 Additional outcomes

Two additional outcomes are used in the mechanism heterogeneity analysis.

Smoking behaviour. Ever smoking is measured at age 17 as a binary indicator of whether the child has ever smoked regularly. The trait-matched standard smoking PGI is constructed from GWAS summary statistics for ever smoking reported in the genome-wide association study of risk tolerance and risky behaviours by Karlsson Linnér et al. (2019). The corresponding DGE smoking PGI is constructed from the family-based GWAS weights of Tan et al. (2026).

Child behavioural and emotional difficulties. The Strengths and Difficulties Questionnaire (SDQ) total difficulties score is a 25-item parent-reported screening instrument for child behavioural and emotional problems, widely used in developmental and population studies. The total difficulties score sums four subscales (emotional symptoms, conduct problems, hyperactivity-inattention, and peer relationship problems), with higher scores indicating greater difficulties. SDQ total difficulties are measured at age 17. The trait-matched standard subjective wellbeing (SWB) PGI is constructed from Okbay et al. (2016); the corresponding DGE SWB PGI is constructed from Tan et al. (2026).

Because both ever smoking and SDQ total difficulties are measured only at age 17, the mechanism heterogeneity analyses use the age 17 cross-section rather than the pooled panel.

4.6 Control variables

All specifications from Column (3) onward include a standard set of controls: sex, cohort indicators, and the first ten genetic principal components (PCs). Cohort indicators capture the stratified sampling design of the MCS. Genetic principal components are computed from the genotype data and absorb broad ancestry differences that could otherwise confound genetic associations with outcomes.

Data limitations. The MCS genetic sample does not include detailed measures of parental health behaviours, dietary provision, or household food environments. The "family channel" identified in this paper is therefore inferred from the statistical contrast between the standard and DGE PGIs and from parental PGI controls, rather than from direct observation of household mechanisms. This limitation is discussed further in the interpretation of results.

4.7 Descriptive statistics

Table 4 reports summary statistics for the estimation sample. The sample comprises 5,446 children contributing 25,997 child–wave observations across ages 5, 7, 11, 14, and 17. Mean BMI increases from 16.34 at age 5 to 23.41 at age 17, with within-wave dispersion widening over time. The standard BMI PGI has near-unit variance by construction, while the DGE PGI—which retains only directly inherited variation—has slightly lower dispersion. The sample is approximately balanced by sex (51% female) and skews modestly toward higher-SES households, consistent with the genotype-consent attrition pattern documented for the MCS.

5 Estimation

The specifications below implement the H1–H2 discrimination strategy formalised in Part III. The core design element is the comparison between the standard PGI × SES interaction and the DGE PGI × SES interaction across a stepwise sequence of models that progressively incorporate parental genetic background, holding the outcome, sample, and controls fixed throughout.

5.1 Baseline associations

Let BMI_{it} denote the body mass index of child i observed at wave t. Let PGI_i denote either the standard child BMI polygenic index or the DGE polygenic index. Let SES_i denote baseline socioeconomic status. All specifications include wave fixed effects δ_t, which absorb systematic age-related differences in BMI levels, and a vector of baseline controls X_i consisting of sex, cohort indicators, and the first ten genetic principal components to account for residual population stratification.

We begin by estimating bivariate benchmarks that summarise unconditional associations:

Table 4: Descriptive statistics

	N	**Mean**	**SD**	**Min**	**Max**
Polygenic indices					
Standard BMI PGI (child)	5,446	−0.03	1.07	−4.25	3.72
DGE BMI PGI (child)	5,446	0.01	1.00	−2.81	4.58
Mother BMI PGI	5,446	−0.05	1.03	−3.82	3.84
Father BMI PGI	5,446	−0.05	0.83	−3.97	4.63
Socioeconomic status and demographics					
SES quintile (1–5)	5,446	3.18	1.36	1	5
Female	5,446	0.51	0.50	0	1
Waves per child	5,446	4.77	0.54	1	5
BMI by survey wave					
Age 5	5,231	16.34	1.80	11.64	36.71
Age 7	5,139	16.61	2.26	10.11	32.17
Age 11	5,224	19.20	3.52	9.48	40.30
Age 14	5,314	21.54	4.14	12.53	46.49
Age 17	5,089	23.41	4.73	14.10	49.38

Notes: The estimation sample includes children with non-missing BMI (> 0), genotype data, parental PGIs (observed or imputed), SES quintile, and control variables. Polygenic indices, SES, and demographic variables are reported at the child level (one observation per child). All polygenic indices are standardised to mean zero and unit variance in the full genotyped sample. SES quintiles are based on modified OECD equivalised household income at baseline (1 = most deprived, 5 = most affluent). The BMI panel reports wave-specific statistics; each child contributes up to five BMI observations (mean 4.77 waves).

$$BMI_{it} = \alpha + \beta_G PGI_i + u_{it}, \tag{15}$$
$$BMI_{it} = \alpha + \beta_S SES_i + v_{it}. \tag{16}$$

We then estimate a controlled baseline specification:

$$BMI_{it} = \alpha + \beta_G PGI_i + \beta_S SES_i + \delta_t + X_i'\gamma + \varepsilon_{it}. \tag{17}$$

This specification isolates the association between genetic predisposition and BMI net of socioeconomic background and demographic controls.

5.2 Socioeconomic moderation

To examine whether the genetic association varies across the socioeconomic distribution, we estimate

$$BMI_{it} = \alpha + \beta_G PGI_i + \beta_S SES_i + \beta_{GS}(PGI_i \times SES_i) + \delta_t + X_i'\gamma + \varepsilon_{it}. \tag{18}$$

When SES is centred at the middle quintile, β_G represents the genetic association at the midpoint of the income distribution, and β_{GS} captures how the strength of the genetic association changes as socioeconomic status increases. A negative β_{GS} implies weaker genetic associations at higher SES.

5.3 Conditioning on parental genetic background

As established in Part II, child PGIs are correlated with parental genotypes through inheritance, so the standard PGI bundles direct genetic effects with indirect family-level channels. To absorb these channels and implement the H1–H2 comparison, we extend the model to include maternal and paternal polygenic indices:

$$\begin{aligned} BMI_{it} = \alpha &+ \beta_G PGI_i + \beta_S SES_i + \beta_{GS}(PGI_i \times SES_i) \\ &+ \beta_m PGI_i^m + \beta_p PGI_i^p + \delta_t + X_i'\gamma + \varepsilon_{it}. \end{aligned} \tag{19}$$

We then estimate a more flexible specification that additionally includes interactions between the child polygenic index and parental polygenic indices, and between parental polygenic indices and socioeconomic status:

$$\begin{aligned} BMI_{it} = \alpha &+ \beta_G PGI_i + \beta_S SES_i + \beta_{GS}(PGI_i \times SES_i) \\ &+ \beta_m PGI_i^m + \beta_p PGI_i^p + \phi_{mc}(PGI_i^m \times PGI_i) + \phi_{pc}(PGI_i^p \times PGI_i) \\ &+ \phi_{ms}(PGI_i^m \times SES_i) + \phi_{ps}(PGI_i^p \times SES_i) + \delta_t + X_i'\gamma + \varepsilon_{it}. \end{aligned} \tag{20}$$

This flexible interaction structure ensures that estimated child-level moderation is not mechanically driven by restrictions imposed on parental genetic effects. Equation (19) corresponds to Column (5) in the results tables; equation (20) corresponds to Column (7).

5.4 Age heterogeneity

The panel structure allows moderation to vary across developmental stages. Let $\mathbf{1}\{t = s\}$ denote an indicator for wave s. We estimate specifications that include the full set of two-way interactions between the polygenic index and wave indicators, between SES and wave indicators, and the corresponding three-way interaction. This yields wave-specific PGI × SES slopes without imposing that moderation is constant across ages.

5.5 Nonlinearity in SES

To allow for nonlinear moderation, we replace the continuous SES measure with quintile indicators. Let $\mathbf{1}\{SES_i = q\}$ denote membership of income quintile q. We estimate

$$\begin{aligned} BMI_{it} = \alpha + \beta_G PGI_i + \sum_q \pi_q \mathbf{1}\{SES_i = q\} + \sum_q \lambda_q (PGI_i \times \mathbf{1}\{SES_i = q\}) \\ + \beta_m PGI_i^m + \beta_p PGI_i^p + \delta_t + X_i' \gamma + \varepsilon_{it}. \end{aligned} \tag{21}$$

This yields group-specific genetic slopes and allows heterogeneity patterns to be presented without relying on linearity assumptions.

5.6 Statistical inference

Because each child contributes multiple observations across waves while the polygenic indices and baseline SES are time-invariant, residuals may be correlated within child. Our preferred inference therefore uses heteroskedasticity-robust standard errors clustered at the child level, which allow arbitrary serial correlation in the error term across waves for the same individual. For transparency, we also report conventional OLS standard errors in selected specifications; because these treat repeated observations as independent, they may understate sampling uncertainty.

5.7 Design validation

Three design features support the validity of the H1–H2 discrimination test.

The DGE comparison. The DGE PGI is constructed from family-based GWAS estimates that condition on parental genotypes, absorbing indirect genetic effects and stratification confounding into the parental controls. Because the DGE and standard PGIs are applied to the same sample with the same outcome and controls, the difference in their interaction with SES is a direct empirical test of whether family-level channels account for the observed heterogeneity. No additional assumptions about selection or functional form are required beyond those already embedded in the main regressions.

Trio sample robustness. A concern with the design is that Mendelian-imputed parental genotypes (Young et al., 2022) may introduce measurement error into parental PGIs, potentially attenuating coefficients and understating the DGE comparison. We address this by restricting the sample to families in which child, maternal, and paternal genotypes are all directly observed—that is, excluding any child whose parental genotypes were imputed from siblings or other relatives. If imputation biases the main analysis, restricting to observed trios should either weaken or reverse the central pattern. In practice, the opposite occurs: in the trio subsample ($N = 2{,}707$ children, 13,057 child–wave observations), the standard child PGI $\times$ SES coefficient strengthens to -0.129 and is significant at the 5 percent level, while the DGE PGI $\times$ SES coefficient remains indistinguishable from zero. Both findings are reported in Table 12. This confirms that imputation, if anything, introduces conservative attenuation rather than spurious moderation, and that the Standard-vs-DGE contrast is not an artefact of the imputation procedure.

Mechanism heterogeneity across outcomes. We apply the same Standard-vs-DGE comparison to two additional outcomes: smoking behaviour and child behavioural and emotional difficulties, measured by the Strengths and Difficulties Questionnaire (SDQ) total difficulties score.[2] These outcomes are genetically and socioeconomically patterned but involve different biological and social pathways. The cross-outcome comparison serves two purposes. First, if the Standard PGI $\times$ SES interaction is zero for an alternative outcome, it demonstrates that the moderation pattern is not a mechanical feature of interacting any polygenic predictor with any socially patterned outcome. Second, comparing the direction and magnitude of the Standard-vs-DGE contrast across traits reveals whether the mechanism generating moderation is trait-specific or generic.

6 Results

6.1 Baseline gradients in BMI by genetics and socioeconomic status

Table 6 reports a stepwise sequence of pooled specifications for the Standard child BMI PGI (Panel A) and the child DGE PGI (Panel B). All models include wave fixed effects and the standard control set, and inference is based on standard errors clustered at the child level.

Panel A: Standard PGI. Column (1) establishes that a one-standard-deviation increase in the Standard BMI PGI is associated with an increase of approximately 0.9 BMI units. Column (2) confirms the SES gradient: each one-quintile increase in income rank corresponds to a reduction of 0.30 BMI units. In Column (3), both predictors enter jointly with stable coefficients, indicating that the genetic and socioeconomic gradients are largely independent.

Column (4) introduces the child PGI $\times$ SES interaction. The coefficient is negative (-0.042), implying that the PGI–BMI association is steeper at lower income rank: the implied PGI slope ranges from 1.04 in the lowest quintile to 0.87 in the highest, a bottom-to-top difference of 0.17 BMI units per standard deviation. This coefficient is not statistically significant at conventional levels, but the sign and magnitude are stable as additional controls are introduced.

Columns (5) through (7) progressively incorporate parental PGIs and their interactions. In Column (7), the most complete specification, the child PGI $\times$ SES coefficient reaches -0.070 and is statistically significant at the 5 percent level. The implied PGI slope is 1.04 in quintile 1 and 0.76 in quintile 5, a bottom-to-top difference of 0.28 BMI units per standard deviation. The father PGI $\times$ SES coefficient in this specification is positive and significant at the 5 percent level

[2] The SDQ is a 25-item parent-reported screening instrument for child behavioural and emotional problems, widely used in developmental and population studies. The total difficulties score sums four subscales (emotional symptoms, conduct problems, hyperactivity-inattention, and peer relationship problems), with higher scores indicating greater difficulties.

(0.096), indicating that the paternal genetic contribution to child BMI is larger at lower income rank—directly consistent with H2, where parental genetic channels are more consequential when household resources are scarcer. Across Columns (4) through (7), the negative sign of the child interaction is consistent, and its magnitude and precision increase as the specification more fully accounts for parental genetic pathways.

Panel B: DGE PGI. The DGE PGI isolates the child's directly inherited genetic variation, purged of parental and stratification channels. In Column (3), the DGE PGI–BMI association is 0.142 BMI units per standard deviation—substantially smaller than the standard PGI coefficient, consistent with the standard PGI capturing additional family-level genetic signal. When parental PGIs are introduced in Column (5), the maternal and paternal coefficients are 0.584 and 0.428 respectively, confirming that parental genotypes independently predict child BMI above and beyond the child's own direct genetic effects.

The key result is the interaction. Across Columns (4) through (7), the DGE PGI $\times$ SES coefficient is close to zero and not statistically significant in any specification. The implied DGE PGI slopes at the bottom and top of the income distribution are correspondingly similar, with no systematic ordering by socioeconomic status.

The contrast between Panels A and B is the central result. The standard PGI shows a negative income gradient in its association with BMI that grows in magnitude and precision as parental genetic pathways are accounted for; the DGE PGI shows no corresponding gradient. This divergence is the predicted signature of H2: SES moderation reflects family-level channels—parental behaviours, household environments, and related pathways—rather than environmental amplification of children's own alleles. The subsections below probe the robustness and specificity of this finding across alternative SES measures, developmental stages, nonlinear income specifications, and alternative outcomes.

6.2 Robustness to SES functional form: standardised log equivalised income

The baseline specification uses income quintile rank, which imposes a linear mapping from ordinal position to the PGI slope. If H1 were correct, the DGE PGI $\times$ SES interaction should emerge regardless of how SES is parameterised; a result that appears only under one functional form would be harder to interpret as genuine moderation.

Table 7 repeats the same stepwise sequence but replaces income rank with standardised log equivalised income. In Panel A, the Standard PGI coefficient is stable across columns and the income main effect is negative. The Standard PGI $\times$ income interaction is negative in sign but not statistically significant at conventional levels, so the linear log-income specification does not yield a precisely estimated SES gradient in the PGI slope. In Panel B, the DGE PGI $\times$ income interaction remains close to zero. The qualitative pattern—negative but imprecise for the standard predictor, near zero for the DGE predictor—is consistent with the baseline results.

6.3 Age heterogeneity in SES moderation

Under H1, direct-effect moderation should operate wherever biological amplification occurs—potentially across the entire developmental window from childhood to adolescence. Under H2, moderation driven by household environments may concentrate at ages when parental influence over diet and activity is most consequential.

Table 8 allows the child PGI slope, the SES gradient, and their interaction to vary by age wave using interactions with age indicators. The reported coefficients imply age-specific PGI slopes at any SES value and SES-specific PGI slopes at any age wave, without imposing that moderation is constant across development.

For the Standard PGI, the age interactions imply an increase in the PGI slope from age 5 to ages 7, 11, 14, and 17. The SES gradient also becomes more negative at older ages. The three-way interaction at age 17 is negative and statistically significant (-0.093, $p = 0.018$), implying that the socioeconomic difference in the Standard PGI slope is larger in late adolescence than in early childhood. At earlier ages, the corresponding three-way terms are small and not statistically significant, so the age-heterogeneity specification locates the SES-graded difference in the Standard PGI slope primarily at age 17. Figure 2 visualises these implied slopes: Panel A (Standard PGI) shows the three SES-specific lines diverging with age, while Panel B (DGE PGI) shows them bunched together near zero at every age.

For the DGE PGI, the age interactions are small and the three-way terms do not imply a systematic ordering of the DGE PGI slope by income rank at any age.

6.4 Nonlinear SES heterogeneity

The linear interaction specification constrains the PGI slope to change at a constant rate across the income distribution. Relaxing this assumption allows the data to reveal whether moderation is concentrated at the bottom of the distribution —as a resource-constraint mechanism would predict—or distributed more evenly.

Table 9 replaces the linear SES interaction with interactions between the child PGI and SES quintile indicators, with quintile 1 as the reference group. This specification yields quintile-specific PGI slopes and therefore provides an interpretation based on differences in PGI slopes across discrete points of the income distribution.

For the Standard PGI, the estimated PGI slope in quintile 1 equals 1.049 BMI units per standard deviation. The implied slopes in quintiles 2 to 5 are obtained by adding the corresponding interaction coefficients to the quintile 1 slope. Under this parameterisation, the implied slope in quintile 5 equals 0.728, and the implied difference between quintile 1 and quintile 5 slopes equals 0.321 BMI units per standard deviation. Figure 3 visualises the full quintile profile: Panel A shows the steady decline in the Standard PGI slope across quintiles, while Panel B shows a flat DGE PGI slope centred near zero at every quintile.

For the DGE PGI, the implied quintile-specific slopes do not exhibit a monotone ordering from quintile 1 to quintile 5, confirming that there is no systematic socioeconomic gradient in the DGE PGI–BMI association.

6.5 Mechanism heterogeneity across outcomes

The DGE comparison establishes that SES moderation of the Standard BMI PGI reflects family-level channels rather than children's direct genetic effects. A complementary question is whether this pattern is specific to BMI or whether the same Standard-vs-DGE contrast appears broadly across outcomes. If moderation were a mechanical feature of regressing any polygenic predictor on a socially patterned outcome, the same signature would replicate everywhere. If instead the mechanism varies across outcomes in theoretically interpretable ways, the cross-outcome comparison itself becomes informative about the biology and sociology of gene–environment interplay. I examine two additional outcomes—smoking behaviour and child behavioural difficulties—each with its own trait-matched standard PGI and DGE PGI. The standard smoking PGI is constructed from GWAS summary statistics for ever smoking (Karlsson Linnér et al., 2019);[3] the standard subjective wellbeing (SWB) PGI is constructed from Okbay et al. (2016).[4] Each outcome's DGE PGI is constructed from the corresponding family-based GWAS weights (Tan et al., 2026). Because both ever smoking and SDQ total difficulties are measured only at age 17, these analyses use the age 17 cross-section rather than the pooled panel, so each individual contributes a single observation.

The three outcomes together yield a 3×2 pattern of Standard-vs-DGE contrasts that is both internally consistent and substantively meaningful:

Smoking behaviour. Table 10 reports the stepwise models for ever smoking. Panel A shows that the Standard smoking PGI is positively associated with ever smoking and there is a strong SES gradient, but the child smoking PGI $\times$ SES_c interaction is close to zero across all specifications. Panel B shows the same for the DGE smoking PGI: neither the standard nor the direct-effects predictor exhibits meaningful SES moderation. For smoking, genetic predisposition—whether measured through the composite standard PGI or the direct-effects PGI—does not vary systematically in its association with the outcome across the income distribution. This establishes that socioeconomic moderation is not mechanically induced whenever a PGI predicts a socially patterned outcome.

Child behavioural and mental health difficulties. Table 11 reports results for SDQ total difficulties using the SWB polygenic predictors described above. Panel A shows that the Standard SWB PGI is negatively associated with SDQ difficulties and there is a strong SES gradient, but the PGI $\times$ SES interaction is small and not statistically significant—the same null as the standard smoking PGI. Panel B shows a directionally different pattern: the DGE SWB PGI $\times$ SES interaction is negative in all specifications (point estimates between -0.066 and -0.077), but not statistically significant at conventional levels. The sign is consistent with children whose direct genetic endowment for wellbeing is higher showing a more protective association with behavioural difficulties at lower income rank, but the smaller cross-sectional sample at age 17 does not provide sufficient precision to confirm this pattern. The contrast with the Standard PGI is nevertheless noteworthy: moderation point estimates are concentrated in the DGE predictor rather than the Standard predictor, the reverse of the BMI pattern.

Cross-outcome pattern and its interpretation. Table 5 summarises the three-outcome pattern. For BMI, moderation is driven by family-level channels captured in the composite standard

[3] The ever-smoking GWAS captures genetic variants associated with whether an individual has ever smoked regularly. A higher smoking PGI indicates greater genetic propensity for smoking initiation.

[4] The SWB GWAS captures genetic variants associated with self-reported life satisfaction and positive affect. A higher SWB PGI indicates greater genetic propensity for subjective wellbeing; negative associations with SDQ total difficulties therefore imply fewer behavioural problems among children with higher wellbeing-related genetic endowments.

PGI but absorbed when conditioning on parental genotypes (consistent with H2). For smoking, neither channel generates meaningful SES moderation. For mental health, the point estimates are suggestive of the opposite pattern—moderation concentrated in children's direct genetic endowment rather than in the standard predictor—but the effect does not reach statistical significance in the age 17 cross-section.

The cross-outcome pattern is nevertheless informative. The absence of moderation for smoking confirms that a significant PGI × SES interaction is not a mechanical property of interacting any polygenic predictor with any socially patterned outcome. The directional contrast for SDQ—where the DGE rather than the Standard predictor shows the larger interaction—is consistent with direct-effect moderation (H1), even though the precision of the cross-sectional estimate does not permit a definitive conclusion. For BMI, the clear Standard-vs-DGE contrast across all specifications and functional forms provides the strongest evidence, firmly supporting family-channel moderation (H2).

Table 5: Cross-outcome pattern of Standard-vs-DGE moderation contrasts

Outcome	**Standard PGI × SES**	**DGE PGI × SES**	**Dominant mechanism**
BMI	Negative (sig. in Col. 7)	Near zero	H2: family channels
Smoking	Near zero	Near zero	No moderation
SDQ difficulties	Near zero	Negative (n.s.)	Suggestive of H1

Notes: Each outcome uses a trait-matched standard PGI and DGE PGI estimated from the same stepwise specifications. Smoking and SDQ analyses use the age 17 cross-section (single observation per child); BMI uses the pooled age 5–17 panel. "Sig." denotes statistically significant; "n.s." denotes not statistically significant at conventional levels. The SDQ point estimates are directionally consistent with H1 (direct-effect moderation) but imprecisely estimated in the cross-sectional sample.

6.6 Robustness to imputed parental genotypes: the trio sample

A potential concern with the main specification is that a non-trivial share of parental polygenic indices is imputed rather than directly observed (Young et al., 2022), raising the possibility that measurement error or selective missingness in parental genotypes could bias estimates of genetic effects and gene–environment interactions. To assess the relevance of this concern, we re-estimate the main specification on a restricted trio sample in which both paternal and maternal genotypes are directly observed—no imputation of any kind—yielding 2,707 unique children in Panel A (Standard PGI) and 2,697 in Panel B (DGE PGI). For Panel B, the child, mother, and father DGE PGIs are recomputed from the family-based GWAS weights on this real-trio subsample alone, so neither the analysis sample nor the DGE PGI relies on imputed parental genotypes. Table 12 reports the results.

Restricting the analysis to the trio sample does not weaken the central pattern; it strengthens it. In Panel A, the child Standard PGI × SES interaction in Column (7) is -0.129, significant at the 5 percent level ($p = 0.023$). This is roughly twice as large in magnitude as the corresponding full-sample estimate of -0.070, despite the trio sample being half the size. The implied PGI slope is 1.10 in the lowest income quintile and 0.58 in the highest, a bottom-to-top difference of 0.52 BMI units per standard deviation. In Panel B, the DGE PGI × SES interaction in Column (7) is $+0.005$ ($p = 0.95$)—point-estimate indistinguishable from zero, and an even cleaner null than in the full sample. The maternal and paternal trio DGE PGIs are also not individually significant as interactions with SES.

Two implications follow. First, the Standard-vs-DGE contrast that identifies H2 is not an artefact of Mendelian imputation. When the imputation step is entirely removed from both the sample definition and the construction of the PGIs, the Standard PGI retains a significant negative interaction with SES and the DGE PGI retains a null interaction. Second, imputation introduces conservative attenuation rather than spurious moderation: the magnitude of the Standard PGI × SES coefficient nearly doubles when imputation is removed, consistent with classical measurement error in the imputed parental genotypes biasing the full-sample estimate toward zero.

Finally, the Standard PGI main effect is also smaller in the trio sample (Column (7): 0.843 versus 0.900 in the full sample), and the DGE PGI main effect is larger and significant (0.230 versus 0.095). Together, these patterns suggest that the real-trio construction produces cleaner parental genetic controls than the imputed-parent analysis, and that our main conclusions regarding the unequal expression of genetic risk across the socioeconomic distribution are conservative with respect to the imputation procedure.

Selection into the trio subsample. A natural concern is whether the trio subsample is a random subset of the full estimation sample. Table A2 in the appendix reports descriptive comparisons of the 2,707 children in the real-trio subsample and the 2,739 children whose inclusion in the main analysis required Mendelian imputation of at least one parental genotype. The two groups differ systematically on socioeconomic dimensions: trio children come from higher-income households (mean quintile 3.45 versus 2.91, $p < 0.001$) with correspondingly lower Standard BMI PGIs (-0.085 versus $+0.033$, $p < 0.001$) and lower mean BMI (19.27 versus 19.65, $p < 0.001$). This pattern is consistent with the well-documented selection of more stable, higher-income families into genotype consent for both parents, itself an instance of the broader socioeconomic gradient in cohort-study participation and consent.

Crucially, however, the DGE BMI PGI—the child's direct-effect genetic predictor used in Panel B of the trio table—is balanced between the two groups (mean -0.008 versus $+0.018$, $p = 0.33$). Selection into the trio subsample is therefore on the socioeconomic dimension, not on the direct-effect genetic dimension that identifies H1. Because the trio sample systematically excludes the most disadvantaged families (the quintile 1 share drops from 20.7% in the imputed-parent subsample to 8.8% in the trio subsample), it compresses the very range where H2 predicts the strongest moderation; yet the estimated Standard PGI × SES interaction nearly doubles in magnitude (-0.129 versus -0.070). The strengthening of the interaction under an SES-compressed sample—together with the balanced DGE PGI—rules out the interpretation that selection into the trio subsample generates the H2 pattern.

6.7 Mechanism assessment

The results across Sections 6.1–6.4 provide consistent evidence in favour of H2 over H1. The key discriminating test—the DGE PGI × SES interaction—is close to zero and not statistically significant in any specification or functional form for SES. By contrast, the standard PGI × SES interaction is negative and, in the most complete specification, implies a bottom-to-top difference in the PGI slope of 0.28 BMI units per standard deviation. The divergence between the two predictors—moderation present for the standard PGI, absent for the DGE PGI—is the predicted signature of H2: it arises when family-level channels—including household environments and parental behaviours—rather than children's own alleles, are the source of socioeconomic heterogeneity.

Three features of the evidence strengthen this assessment. First, the pattern is not confined to a single specification: it holds across linear income rank, income quintile, and log income parameterisations. Second, it is specific to BMI: smoking shows no interaction for either predictor, and the SDQ point estimates are directionally consistent with DGE-based moderation (H1) rather than family-channel moderation (H2)—suggesting that the finding reflects something particular to the BMI–family-environment pathway rather than a general property of this sample or modelling approach. Third, the trio sample results rule out the main mechanical alternative: imputed paternal genotypes attenuate rather than generate the moderation pattern, so the finding is conservative with respect to measurement error.

The age heterogeneity and quintile results add texture without changing the interpretation. The standard PGI heterogeneity is concentrated in late adolescence and at the lower end of the income distribution, consistent with household dietary environments and resource constraints being most consequential during adolescent weight gain. These patterns are descriptive conditional on the DGE comparison having established the channel.

One qualification is warranted. The near-zero DGE PGI $\times$ SES interaction establishes that children's own directly-inherited alleles do not exhibit systematic socioeconomic moderation at the precision available in this sample. It does not rule out a smaller H1 component beneath current detectability: the DGE PGI is itself a noisy construct, and if its weights understate direct genetic effects, the test is conservative. The evidence supports H2 as the dominant explanation for the observed standard PGI heterogeneity, while leaving open the possibility of residual direct-effect moderation that cannot yet be measured precisely.

Table 6: Child PGIs, socioeconomic status, and BMI (Clustered)

	Dependent variable: BMI_{it}						
	(1) PGI	(2) SES	(3) + Controls	(4) + PGI×SES	(5) + Parents	(6) + Parent×Child	(7) + Parent×SES
Panel A: Standard PGI							
Child PGI	0.921***		0.947***	0.956***	0.883***	0.889***	0.900***
	(0.036)		(0.038)	(0.039)	(0.049)	(0.050)	(0.051)
SES_c		-0.303***	-0.189***	-0.189***	-0.181***	-0.180***	-0.176***
		(0.030)	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)
Mother PGI					0.148***	0.151***	0.145***
					(0.045)	(0.046)	(0.047)
Father PGI					0.013	0.011	-0.027
					(0.048)	(0.049)	(0.055)
Child PGI × SES_c				-0.042	-0.040	-0.028	-0.070**
				(0.028)	(0.027)	(0.028)	(0.036)
Mother PGI × Child PGI						0.070**	0.070**
						(0.034)	(0.035)
Father PGI × Child PGI						0.016	0.028
						(0.037)	(0.037)
Mother PGI × SES_c							0.030
							(0.035)
Father PGI × SES_c							0.096**
							(0.040)
Observations	5,446	5,446	5,446	5,446	5,446	5,446	5,446
R^2	0.049	0.009	0.442	0.442	0.443	0.443	0.444
Panel B: DGE PGI							
Child DGE PGI	0.092**		0.142**	0.141**	0.096	0.096	0.095
	(0.039)		(0.066)	(0.068)	(0.066)	(0.067)	(0.067)
SES_c		-0.303***	-0.289***	-0.289***	-0.210***	-0.211***	-0.210***
		(0.030)	(0.030)	(0.030)	(0.030)	(0.030)	(0.030)
Mother PGI					0.584***	0.584***	0.590***
					(0.040)	(0.040)	(0.041)
Father PGI					0.428***	0.429***	0.406***
					(0.044)	(0.044)	(0.049)
DGE PGI × SES_c				0.004	-0.005	-0.009	-0.008
				(0.030)	(0.030)	(0.031)	(0.031)
Mother PGI × DGE PGI						-0.046	-0.048
						(0.041)	(0.041)
Father PGI × DGE PGI						0.043	0.046
						(0.042)	(0.042)
Mother PGI × SES_c							-0.028
							(0.030)
Father PGI × SES_c							0.049
							(0.036)
Observations	5,446	5,446	5,446	5,446	5,446	5,446	5,446
R^2	0.000	0.009	0.397	0.397	0.419	0.419	0.419

Notes: Columns (1) and (2) are bivariate regressions. Columns (3) to (7) additionally include age fixed effects, sex, cohort indicators, and the first ten genetic principal components. Socioeconomic status is the OECD equivalised income quintile score centred at 3, so $SES_c = 0$ corresponds to quintile 3. Standard errors in parentheses are clustered at the child level. Significance: * 10%, ** 5%, *** 1%.

Table 7: Child PGIs, equivalised household income, and BMI (Clustered)

	Dependent variable: BMI_{it}						
	(1) PGI	(2) Income	(3) + Controls	(4) + PGI×Income	(5) + Parents	(6) + Parent×Child	(7) + Parent×Income
Panel A: Standard PGI							
Child PGI	0.923***		0.955***	0.964***	0.890***	0.896***	0.906***
	(0.036)		(0.038)	(0.040)	(0.050)	(0.050)	(0.051)
SES_use		-0.391***	-0.233***	-0.232***	-0.222***	-0.223***	-0.216***
		(0.043)	(0.041)	(0.041)	(0.041)	(0.041)	(0.041)
Mother PGI					0.150***	0.153***	0.147***
					(0.045)	(0.046)	(0.047)
Father PGI					0.009	0.007	-0.031
					(0.048)	(0.049)	(0.055)
Child PGI × SES_use				-0.046	-0.043	-0.027	-0.085*
				(0.038)	(0.038)	(0.039)	(0.051)
Mother PGI × Child PGI						0.072**	0.072**
						(0.035)	(0.035)
Father PGI × Child PGI						0.019	0.030
						(0.037)	(0.037)
Mother PGI × SES_use							0.040
							(0.049)
Father PGI × SES_use							0.131**
							(0.057)
Observations	5,406	5,406	5,406	5,406	5,406	5,406	5,406
Panel B: DGE PGI							
Child DGE PGI	0.093**		0.154**	0.145**	0.098	0.099	0.097
	(0.039)		(0.067)	(0.069)	(0.067)	(0.067)	(0.067)
SES_use		-0.391***	-0.371***	-0.371***	-0.265***	-0.266***	-0.263***
		(0.043)	(0.043)	(0.043)	(0.043)	(0.043)	(0.043)
Mother PGI					0.592***	0.592***	0.597***
					(0.040)	(0.040)	(0.042)
Father PGI					0.429***	0.430***	0.404***
					(0.044)	(0.044)	(0.050)
DGE PGI × SES_use				0.044	0.032	0.028	0.029
				(0.045)	(0.044)	(0.045)	(0.045)
Mother PGI × DGE PGI						-0.040	-0.041
						(0.042)	(0.042)
Father PGI × DGE PGI						0.046	0.049
						(0.042)	(0.043)
Mother PGI × SES_use							-0.031
							(0.042)
Father PGI × SES_use							0.077
							(0.050)
Observations	5,406	5,406	5,406	5,406	5,406	5,406	5,406

Notes: Columns (1) and (2) are bivariate regressions. Columns (3) to (7) additionally include age fixed effects, sex, cohort indicators, and the first ten genetic principal components. Socioeconomic status is measured by modified OECD equivalised household income and enters as a standardised log variable. Standard errors in parentheses are clustered at the child level. Significance: * 10%, ** 5%, *** 1%.

Table 8: Age heterogeneity in SES moderation

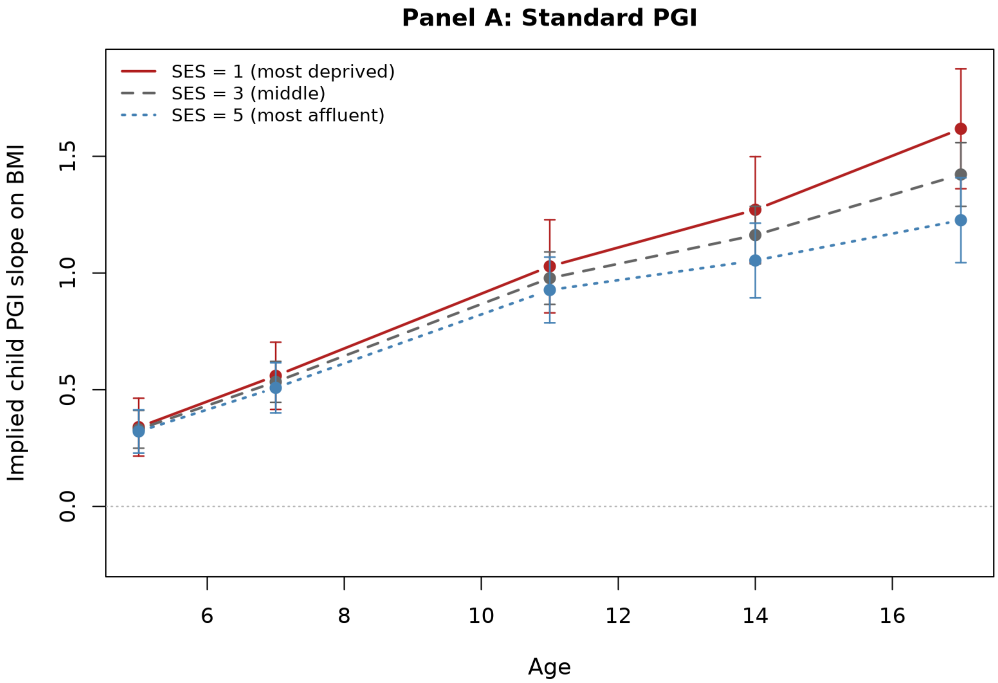
	Standard PGI	DGE PGI
Child PGI (Age 5)	0.330***	0.070
	(0.041)	(0.058)
SES (Age 5)	-0.025	0.020
	(0.019)	(0.020)
Child PGI × Age 7	0.203***	0.015
	(0.023)	(0.023)
Child PGI × Age 11	0.648***	0.039
	(0.039)	(0.041)
Child PGI × Age 14	0.832***	0.054
	(0.047)	(0.050)
Child PGI × Age 17	1.092***	0.023
	(0.056)	(0.062)
SES × Age 7	-0.019	-0.049***
	(0.016)	(0.017)
SES × Age 11	-0.171***	-0.259***
	(0.028)	(0.029)
SES × Age 14	-0.302***	-0.414***
	(0.036)	(0.037)
SES × Age 17	-0.285***	-0.425***
	(0.042)	(0.044)
Child PGI × SES × Age 7	-0.008	-0.001
	(0.015)	(0.017)
Child PGI × SES × Age 11	-0.021	0.004
	(0.026)	(0.029)
Child PGI × SES × Age 14	-0.050	-0.011
	(0.034)	(0.036)
Child PGI × SES × Age 17	-0.093**	0.008
	(0.040)	(0.044)
Mother PGI	0.147***	0.584***
	(0.045)	(0.040)
Father PGI	0.011	0.427***
	(0.048)	(0.044)
Observations	5,446	5,446

Notes: All models include age fixed effects, cohort indicators, sex, and the first ten genetic principal components. Standard errors in parentheses are clustered at the child level. Significance: * 10%, ** 5%, *** 1%.

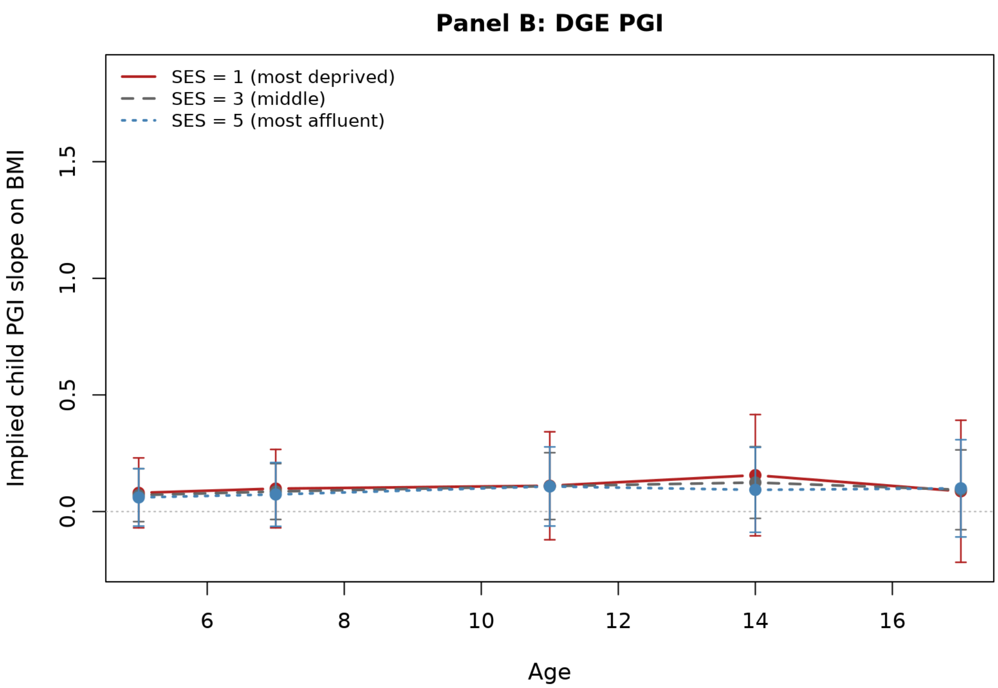
Table 9: Nonlinear SES heterogeneity

	Standard PGI	DGE PGI
Child PGI (Quintile 1 slope)	1.049***	-0.047
	(0.116)	(0.134)
Child PGI × Quintile 2	-0.250*	0.246
	(0.143)	(0.161)
Child PGI × Quintile 3	-0.063	0.259*
	(0.136)	(0.144)
Child PGI × Quintile 4	-0.178	0.038
	(0.131)	(0.137)
Child PGI × Quintile 5	-0.322**	0.131
	(0.131)	(0.141)
SES quintile 2	0.013	-0.142
	(0.148)	(0.156)
SES quintile 3	-0.024	-0.185
	(0.139)	(0.147)
SES quintile 4	-0.264*	-0.409***
	(0.136)	(0.143)
SES quintile 5	-0.711***	-0.881***
	(0.138)	(0.142)
Mother PGI	0.146***	0.585***
	(0.045)	(0.040)
Father PGI	0.013	0.427***
	(0.048)	(0.043)
Observations	5,446	5,446

Notes: SES quintile 1 is the reference group. All models include age fixed effects, cohort indicators, sex, and the first ten genetic principal components. Standard errors in parentheses are clustered at the child level. Significance: * 10%, ** 5%, *** 1%.



(a) Standard PGI



(b) DGE PGI

Figure 2: Age heterogeneity in SES moderation: implied child PGI slopes by age and SES

Notes: The figure reports implied marginal effects of the child PGI on BMI evaluated at SES values 1, 3, and 5 for each age. Implied slopes are computed from a specification that includes wave fixed effects, the full control set, parental PGIs, and interactions that allow the $PGI \times SES$ gradient to vary by wave. Vertical bars indicate 95% confidence intervals based on standard errors clustered at the child level.

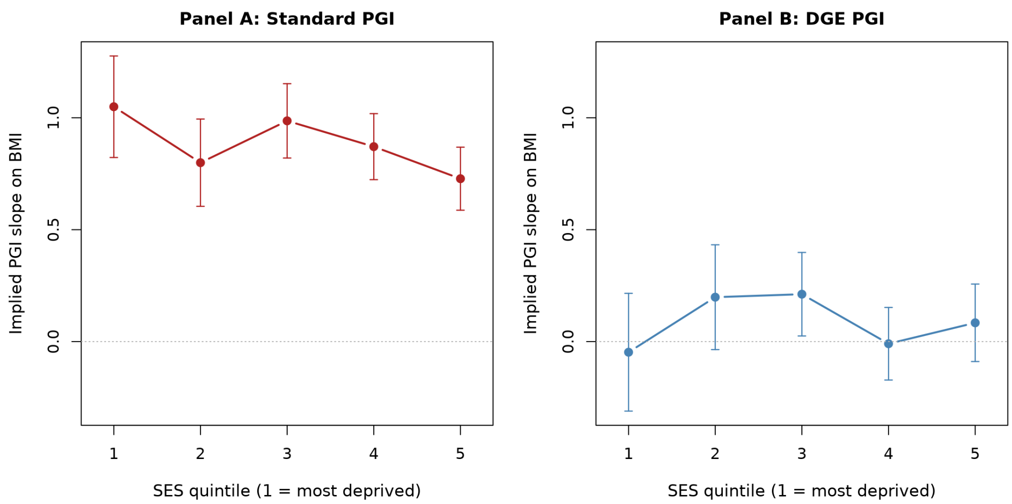


Figure 3: Nonlinear SES heterogeneity: implied child PGI slopes by SES quintile

Notes: The figure reports implied within quintile slopes of BMI on the child PGI from specifications that interact the child PGI with SES quintile indicators. Quintile 1 is the reference group. The implied slope in quintile q equals the PGI coefficient plus the interaction term for quintile q. Vertical bars indicate 95% confidence intervals based on standard errors clustered at the child level. Regressions include wave fixed effects, the full control set, and parental PGIs, and are estimated on the parents available sample for each PGI specification.

Table 10: Mechanism heterogeneity: smoking behaviour on smoking PGIs

	Dependent variable: $EverSmoke_i$						
	(1) PGI	(2) SES_c	(3) + Controls	(4) + PGI×SES_c	(5) + Parents	(6) + Parent×Child	(7) + Parent×SES_c
Panel A: Standard PGI							
Child smoking PGI	0.041***		0.037***	0.038***	0.028***	0.028***	0.028***
	(0.005)		(0.005)	(0.005)	(0.006)	(0.006)	(0.006)
SES_c		-0.034***	-0.033***	-0.033***	-0.032***	-0.032***	-0.032***
		(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Mother PGI					0.013**	0.013**	0.013**
					(0.005)	(0.005)	(0.006)
Father PGI					0.007	0.007	0.006
					(0.005)	(0.005)	(0.006)
Child PGI × SES_c				-0.001	-0.001	-0.000	-0.002
				(0.004)	(0.004)	(0.004)	(0.005)
Mother PGI × Child PGI						0.007*	0.007*
						(0.004)	(0.004)
Father PGI × Child PGI						-0.002	-0.002
						(0.004)	(0.004)
Mother PGI × SES_c							0.002
							(0.004)
Father PGI × SES_c							0.002
							(0.004)
Observations	5,367	5,367	5,367	5,367	5,367	5,367	5,367
R^2	0.015	0.018	0.041	0.041	0.042	0.042	0.043
Panel B: DGE PGI							
Child smoking DGE PGI	0.025***		0.024***	0.025***	0.019***	0.018***	0.018***
	(0.005)		(0.005)	(0.006)	(0.006)	(0.006)	(0.006)
SES_c		-0.016***	-0.019***	-0.019***	-0.016***	-0.016***	-0.016***
		(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Mother PGI					0.023***	0.023***	0.026***
					(0.006)	(0.006)	(0.007)
Father PGI					0.018***	0.018***	0.015**
					(0.006)	(0.006)	(0.007)
Child PGI × SES_c				-0.001	-0.001	0.001	0.001
				(0.004)	(0.004)	(0.004)	(0.004)
Mother PGI × Child PGI						0.001	-0.000
						(0.005)	(0.006)
Father PGI × Child PGI						0.012**	0.012**
						(0.006)	(0.006)
Mother PGI × SES_c							-0.006
							(0.005)
Father PGI × SES_c							0.005
							(0.005)
Observations	2,661	2,661	2,661	2,661	2,661	2,661	2,661
R^2	0.007	0.005	0.022	0.022	0.031	0.032	0.033

Notes: Columns (1) and (2) are bivariate regressions. Columns (3) to (7) additionally include sex, cohort indicators, and the first ten genetic principal components (PC1 to PC10). SES is the OECD equivalised household income quintile score (1 to 5) centred at the middle quintile (so $SES_c = SES - 3$). Child, mother, and father PGIs are standardised (z-scored) once within the common sample used for each panel, and the standardised versions are used throughout. Standard errors in parentheses are heteroskedasticity-robust (HC1) standard errors. Significance: * 10%, ** 5%, *** 1%.

Table 11: Mechanism heterogeneity: SDQ total difficulties on subjective wellbeing PGIs

	Dependent variable: SDQ_i (total difficulties)						
	(1) PGI	(2) SES$_c$	(3) + Controls	(4) + PGI×SES$_c$	(5) + Parents	(6) + Parent×Child	(7) + Parent×SES$_c$
Panel A: Standard PGI							
Child SWB PGI	-0.285***		-0.286***	-0.282***	-0.117	-0.119	-0.121
	(0.065)		(0.066)	(0.069)	(0.088)	(0.088)	(0.088)
SES$_c$		-0.660***	-0.674***	-0.674***	-0.667***	-0.657***	-0.659***
		(0.049)	(0.050)	(0.050)	(0.050)	(0.050)	(0.050)
Mother PGI					-0.235***	-0.234***	-0.250***
					(0.077)	(0.077)	(0.079)
Father PGI					-0.119*	-0.118*	-0.071
					(0.067)	(0.067)	(0.074)
Child PGI × SES$_c$				-0.016	-0.009	0.001	0.009
				(0.049)	(0.049)	(0.050)	(0.065)
Mother PGI × Child PGI						-0.050	-0.057
						(0.055)	(0.055)
Father PGI × Child PGI						-0.144***	-0.139***
						(0.044)	(0.044)
Mother PGI × SES$_c$							0.057
							(0.057)
Father PGI × SES$_c$							-0.095*
							(0.053)
Observations	4,210	4,210	4,210	4,210	4,210	4,210	4,210
R^2	0.005	0.044	0.057	0.057	0.059	0.061	0.062
Panel B: DGE PGI							
Child SWB DGE PGI	-0.008		0.007	0.052	0.087	0.084	0.083
	(0.080)		(0.089)	(0.102)	(0.102)	(0.102)	(0.102)
SES$_c$		-0.488***	-0.494***	-0.492***	-0.485***	-0.482***	-0.487***
		(0.068)	(0.072)	(0.072)	(0.072)	(0.072)	(0.072)
Mother PGI					-0.329***	-0.333***	-0.339***
					(0.080)	(0.080)	(0.089)
Father PGI					-0.197**	-0.203**	-0.140
					(0.080)	(0.080)	(0.089)
Child PGI × SES$_c$				-0.077	-0.073	-0.070	-0.066
				(0.067)	(0.067)	(0.067)	(0.067)
Mother PGI × Child PGI						-0.117	-0.120
						(0.074)	(0.074)
Father PGI × Child PGI						-0.067	-0.072
						(0.077)	(0.078)
Mother PGI × SES$_c$							0.007
							(0.066)
Father PGI × SES$_c$							-0.126**
							(0.062)
Observations	2,330	2,330	2,330	2,330	2,330	2,330	2,330
R^2	0.000	0.024	0.033	0.033	0.042	0.043	0.045

Notes: Columns (1) and (2) are bivariate regressions. Columns (3) to (7) additionally include sex, cohort indicators, and the first ten genetic principal components (PC1 to PC10). SES is the OECD equivalised household income quintile score (1 to 5) centred at the middle quintile (so $SES_c = SES - 3$). Child, mother, and father PGIs are standardised (z-scored) once within the common sample used for each panel, and the standardised versions are used throughout. Standard errors in parentheses are conventional OLS standard errors. Significance: * 10%, ** 5%, *** 1%.

Table 12: Child PGIs, socioeconomic status, and BMI (non-imputed sample)

	Dependent variable: BMI_{it}						
	(1) PGI	(2) SES	(3) + Controls	(4) + PGI×SES	(5) + Parents	(6) + Parent×Child	(7) + Parent×SES
Panel A: Standard PGI							
Child PGI (z)	0.843***		0.883***	0.903***	0.791***	0.791***	0.843***
	(0.050)		(0.050)	(0.058)	(0.075)	(0.075)	(0.080)
SES_c		-0.259***	-0.173***	-0.174***	-0.164***	-0.164***	-0.159***
		(0.043)	(0.043)	(0.043)	(0.043)	(0.043)	(0.043)
Mother PGI (z)					0.191***	0.199***	0.166**
					(0.061)	(0.064)	(0.069)
Father PGI (z)					0.033	0.035	-0.023
					(0.057)	(0.058)	(0.067)
Child PGI $\times SES_c$				-0.037	-0.036	-0.024	-0.129**
				(0.041)	(0.040)	(0.041)	(0.057)
Mother PGI × Child PGI						0.056	0.055
						(0.049)	(0.049)
Father PGI × Child PGI						0.021	0.030
						(0.042)	(0.041)
Mother PGI $\times SES_c$							0.072
							(0.050)
Father PGI $\times SES_c$							0.125***
							(0.048)
Observations	13,057	13,057	13,057	13,057	13,057	13,057	13,057
R^2	0.043	0.006	0.445	0.445	0.446	0.447	0.448
Panel B: DGE PGI							
Child PGI (z)	0.130**		0.207**	0.200**	0.225**	0.226**	0.230**
	(0.055)		(0.081)	(0.089)	(0.103)	(0.103)	(0.108)
SES_c		-0.257***	-0.257***	-0.257***	-0.259***	-0.259***	-0.259***
		(0.043)	(0.045)	(0.045)	(0.045)	(0.045)	(0.045)
Mother PGI (z)					-0.061	-0.057	-0.053
					(0.062)	(0.063)	(0.074)
Father PGI (z)					0.001	-0.002	-0.017
					(0.064)	(0.065)	(0.075)
Child PGI $\times SES_c$				0.014	0.015	0.015	0.005
				(0.047)	(0.047)	(0.047)	(0.069)
Mother PGI × Child PGI						-0.008	-0.007
						(0.044)	(0.044)
Father PGI × Child PGI						0.032	0.032
						(0.048)	(0.048)
Mother PGI $\times SES_c$							-0.006
							(0.053)
Father PGI $\times SES_c$							0.027
							(0.054)
Observations	13,007	13,007	13,007	13,007	13,007	13,007	13,007
R^2	0.001	0.006	0.404	0.404	0.404	0.404	0.404

Notes: Non-imputed trio sample. Children restricted to those with both parents directly genotyped (no snipar imputation of parental genotypes), yielding 2,707 unique children in Panel A (Standard PGI) and 2,697 in Panel B (DGE PGI), contributing 13,057 and 13,007 child–wave observations respectively. Panel B uses child, mother, and father DGE BMI polygenic indices recomputed on the real-trio subsample without any Mendelian imputation, with per-SNP mean-centring performed within this subsample. Columns (1) and (2) are bivariate regressions. Columns (3) to (7) additionally include wave-age fixed effects (ages 5, 7, 11, 14, 17), sex, cohort indicators, and the first ten genetic principal components (PC1 to PC10). SES_c is the OECD equivalised household income quintile score (1 to 5) centred at the middle quintile ($SES_c = SES - 3$). Standard errors in parentheses are clustered at the child level (HC1). Significance: * 10%, ** 5%, *** 1%.

V. Discussion and Conclusion

7 Discussion

7.1 Reinterpreting the PGI × SES literature

A large body of work reports that BMI polygenic indices associate more strongly with body weight among socioeconomically disadvantaged individuals, and interprets this pattern as evidence that adverse environments amplify the expression of genetic predispositions toward obesity (Hüls et al., 2021; Bann et al., 2022). The results of this paper suggest that interpretation requires revision. The Standard-vs-DGE comparison shows that the observed socioeconomic gradient in the PGI–BMI association is driven by family-level channels—not by environmental modification of children's own genetic effects.

This finding speaks directly to the ambiguity identified in the existing literature. Observational cohort studies have reported mixed results: some find that BMI PGIs associate more strongly with weight under disadvantage (Hüls et al., 2021), while others find additive patterns with near-zero interaction terms (de Roo et al., 2023). Our results suggest that these discrepancies may partly reflect differences in the degree to which standard PGIs capture family-level confounding across study designs and cohorts. van den Berg et al. (2025) explicitly noted that their gene–environment estimates "may be capturing the additional exposure driven by the parental genetic predisposition"; Uchikoshi and Conley (2021) called for future work to separate direct genetic mechanisms from genetic nurture in PGI interaction estimates. The present paper provides that separation and confirms that the concern was warranted: the PGI × SES gradient for BMI does not survive when the genetic predictor is restricted to directly inherited effects.

Prior studies that relied on standard population-based PGIs could not distinguish between direct-effect and family-channel moderation because their genetic predictor bundles direct, indirect, and confounding components into a single composite (Benjamin et al., 2024; Biroli et al., 2026). Recent quantification of indirect genetic effects supports this concern: Howe et al. (2022) show that within-sibship GWAS estimates for educationally and socially mediated traits shrink by up to 47% relative to population estimates, and van Alten et al. (2025) find that roughly half of intergenerational genetic effects on socioeconomic outcomes operate through genetic nurture rather than direct genetic transmission. The H2 finding implies that what has been described as "genetic sensitivity to disadvantage" in the BMI literature is more accurately characterised as socioeconomic variation in the family and household environments through which parental genetic background operates.

A potential objection is that BMI shows relatively limited shrinkage in within-sibship GWAS estimates compared to education or cognitive ability (Howe et al., 2022), suggesting that standard and DGE BMI PGIs should be similar. Yet even a modest indirect genetic effect component can generate differential gene–environment interaction patterns if that component is specifically correlated with SES-related household environments. The indirect channel need not be large in the main effect to be consequential in the interaction: family dietary environments and parental health behaviours are precisely the pathways that vary most across the socioeconomic distribution.

This reinterpretation does not diminish the importance of the PGI × SES gradient—it redirects where the action is. The gradient is real: children from disadvantaged families with high-BMI-predisposed parents do gain more weight. But the mechanism runs through what parents do (dietary provision, health behaviours, household organisation) rather than through how children's own alleles express under resource constraint. This distinction matters for both scientific interpretation and intervention design.

7.2 Why mechanism varies across traits

The cross-outcome heterogeneity—H2 for BMI, suggestive of H1 for SDQ, null for smoking—is not incidental; it reflects substantive differences in how genetic predispositions translate into outcomes across the socioeconomic distribution.

For BMI, weight is shaped heavily by the household food environment, which is under parental control and directly constrained by household income (Autret and Bekelman, 2024). Parents who carry BMI-increasing alleles tend to create home dietary environments that raise child weight independently of which alleles the child inherited (Kong et al., 2018; Meddens et al., 2021); under disadvantage, these parental channels are less buffered by alternative food options and health resources, generating the family-channel moderation pattern. Breinholt and Conley (2023) provide direct evidence for a related mechanism: parental cognitively stimulating investments respond to child genotype, and this response is socioeconomically stratified—middle-SES parents reinforce genetic advantages while college-educated parents maintain high investment regardless of child genotype.

For child behavioural and emotional difficulties (SDQ), the relevant genetic pathways—cognitive regulation, emotional reactivity, stress response—operate more directly through the child's own neurodevelopmental endowment. Marees et al. (2021) show that genetic correlates of socioeconomic status pervasively influence the shared heritability architecture of mental health traits, suggesting that SES-related genetic variation is deeply embedded in mental health phenotypes. The point estimates for the DGE SWB PGI $\times$ SES interaction are consistent with this interpretation: they are negative across specifications, suggesting that household disadvantage may amplify direct genetic differences in behavioural difficulties. However, these estimates do not reach statistical significance, so the evidence for H1 in SDQ remains suggestive rather than conclusive. A larger sample or a longitudinal measure of behavioural difficulties would be needed to confirm whether direct-effect moderation operates for mental health outcomes.

For smoking, neither the standard nor the DGE predictor exhibits meaningful SES moderation. This null result is informative precisely because it rules out a trivial explanation: it confirms that socioeconomic moderation is not a mechanical property of interacting any PGI with any socially patterned outcome. The absence of moderation for smoking contrasts with the clear pattern for BMI and the directional evidence for SDQ, and may reflect the fact that adolescent smoking initiation operates through pathways—peer influence, experimentation, access—that do not scale with household income in the same gradient-like manner as diet quality or parental health investment.

This trait-specific pattern has a broader implication: gene–environment interaction is not a monolithic phenomenon. Whether socioeconomic disadvantage amplifies genetic predispositions depends on which biological and social pathways connect genotype to phenotype, and these pathways differ fundamentally across traits.

7.3 Policy implications

The standard interpretation of PGI × SES gradients for BMI—that disadvantage "activates" genetic risk, and therefore interventions should target high-genetic-risk children in disadvantaged settings—rests on the H1 reading. The evidence here suggests that this reading is incorrect for BMI. The gradient is better explained by family-level channels that operate through parental genotypes and household environments.

If H2 is the dominant mechanism, then policies aimed at reducing genetically-driven obesity inequality should focus on the household pathways through which family environments affect children's weight—dietary provision, health behaviours, and resource allocation—rather than on screening children for individual genetic risk. Universal interventions that improve household food environments (e.g., subsidised healthy meals, school nutrition programmes, parental nutrition education) would address the family channel directly. By contrast, precision-medicine approaches that stratify children by genetic risk and allocate interventions accordingly would misidentify the operative channel: the children at highest predicted BMI risk are those whose parents carry obesity-predisposing alleles and live in household environments that reinforce genetic predispositions, not necessarily those whose own alleles are most sensitive to environmental conditions.

For mental health, the directional evidence—though not statistically confirmed—points in the opposite direction. The suggestive H1 pattern for SDQ raises the possibility that children's own genetic endowment for wellbeing interacts with household resources. If confirmed in larger samples, interventions that directly support children's cognitive and emotional development under disadvantage—such as school-based mental health programmes or early-childhood enrichment—would target the correct channel.

7.4 Limitations

Several limitations should be noted. First, the near-zero DGE PGI × SES interaction does not rule out a small H1 component beneath the current precision of the DGE weights; the conclusion is that H2 is the dominant explanation, not necessarily the only one. As family-based GWAS samples grow and DGE weight estimates improve (Tan et al., 2026), the test will gain power to detect smaller direct-effect moderation.

Second, the analysis is based on a single UK birth cohort observed from ages 5 to 17. The UK welfare state provides universal healthcare, free school meals for low-income families, and relatively generous child benefits—institutional features that may buffer some household-level channels. Whether the H2 pattern holds in contexts with weaker social safety nets, where household resources play a larger role in shaping children's health environments, is an open empirical question. Existing evidence suggests that gene–environment interaction patterns are sensitive to institutional context: Isungset et al. (2022) find null or very small gene–SES interactions for educational performance in the Norwegian welfare state, while Houmark et al. (2025) document that the direction of gene–SES interactions in Danish schools shifts across developmental stages. The cross-country generalisability of the finding should be tested in cohorts with different welfare regimes and dietary environments.

Third, while paternal imputation does not bias the main findings—the trio sample produces larger, not smaller, estimates of moderation—the imputed paternal PGI remains a noisy proxy, and fully genotyped family designs would sharpen inference on the paternal channel specifically.

Fourth, the current analysis uses LDpred2 Bayesian shrinkage for PGI construction but does not apply ORIV measurement-error correction to the interaction estimates (Becker et al., 2021). Classical measurement error in the PGI attenuates interaction coefficients toward zero, and this attenuation is likely more severe for the DGE PGI than for the standard PGI because family-based GWAS samples are smaller, yielding noisier weight estimates (Alemu et al., 2025). Incorporating ORIV—as demonstrated by Muslimova et al. (2024) and Pereira and van Kippersluis (2025) for education PGIs—would address residual attenuation and provide a sharper estimate of the magnitude of family-channel moderation.

Finally, while the Standard-vs-DGE comparison identifies the channel through which moderation operates, it does not isolate the specific household mechanism within the family channel. The indirect genetic effect component encompasses parental dietary provision, health behaviours, economic decisions, and neighbourhood selection. Decomposing these sub-channels would require richer data on parental behaviours and household environments than are available in the MCS genetic sample.

7.5 Summary and future directions

This paper asks whether the well-documented socioeconomic gradient in the genetic prediction of BMI reflects environmental amplification of children's own genetic effects or family-level channels embedded in standard polygenic indices. Using a comparison between standard and direct genetic effects PGIs—the first, to my knowledge, applied to a PGI $\times$ SES framework—I find consistent evidence that the gradient reflects family-channel moderation (H2) rather than direct-effect moderation (H1). The standard PGI $\times$ SES interaction is negative and grows as parental genetic pathways are accounted for; the DGE PGI $\times$ SES interaction is close to zero across all specifications. The cross-outcome analysis provides supporting evidence: smoking shows no moderation for either predictor, while SDQ point estimates are directionally consistent with direct-effect moderation (H1) concentrated in the DGE predictor. The absence of any moderation for smoking rules out mechanical artefact, and the directional SDQ contrast—though imprecisely estimated—is consistent with the DGE design being sensitive to direct-effect channels when they operate.

The finding reframes an active literature. What has been interpreted as genetic sensitivity to socioeconomic disadvantage—and cited as motivation for genetically-targeted interventions—is better understood as socioeconomic variation in the family environments that parental genotypes help create. For BMI, the policy lever is the household, not the child's genome.

Several directions for future work follow directly from the limitations of the present analysis and the gaps identified in the broader literature. First, the DGE PGI used here is itself an imperfect proxy for direct genetic effects: as family-based GWAS sample sizes grow and estimation precision improves (Tan et al., 2026; Howe et al., 2022), repeating the Standard-vs-DGE comparison with more powerful DGE weights will sharpen the test and potentially detect small residual H1 components that the current analysis cannot rule out. Second, correcting for measurement error in the PGI using instrumental variable methods such as ORIV (Muslimova et al., 2024) would address the concern that attenuation bias in the interaction term may understate both direct-effect and family-channel moderation; this correction requires split-sample PGI construction that was not available for the present study. Third, the family channel identified here is a composite: it includes parental health behaviours, dietary provision, household organisation, and other pathways that covary with parental genotypes. Decomposing this channel—for example, by exploiting asymmetries between maternal and paternal PGI contributions, or by incorporating direct measures of parental behaviours and household food environments—would clarify which specific family mechanisms drive the socioeconomic gradient. Fourth, extending the Standard-vs-DGE comparison to additional traits and cohorts would reveal whether the trait-specific pattern documented here (H2 for BMI, suggestive of H1 for mental health, null for smoking) replicates across institutional contexts and developmental stages, or whether it is specific to the UK cohort and age range studied. Finally, Mostafavi et al. (2020) show that PGI prediction accuracy varies across socioeconomic strata even within an ancestry group; quantifying whether and how this differential accuracy interacts with the Standard-vs-DGE comparison is necessary to fully separate measurement artefacts from genuine biological and social mechanisms.

Appendix

A. Polygenic index construction

This appendix documents the variant- and sample-level processing steps underlying the two BMI polygenic indices used in the main analysis. The construction of imputed parental genotypes, which feeds into the DGE PGI and into the parental controls in the main regressions, is described separately in Appendix 7.5.

Genotype data and pedigree. Child, maternal, and paternal genotypes for the UK Millennium Cohort Study were obtained as per-chromosome genotype filesets in standard binary format. Variant identifiers were harmonised to a common dbSNP 153 reference, duplicate variants were removed, and per-chromosome files were concatenated into a single genome-wide fileset aligned to human genome build GRCh38. A pedigree linking each child to its biological parents was constructed from the MCS family-linkage file, with missing parents flagged explicitly so that they could be either imputed or excluded depending on the analysis.

Standard BMI PGI. SNP weights were derived from the Yengo et al. (2018) BMI GWAS summary statistics, the largest publicly available meta-analysis at the time of construction, combining the GIANT consortium discovery sample with UK Biobank replication. Summary statistics were coordinated with the MCS genotype panel using LDpred v1.0.10 (Vilhjálmsson et al., 2015), restricted to HapMap3 variants, with the 1000 Genomes European sample as the linkage-disequilibrium reference panel and a frequency-discrepancy tolerance of 0.15 between the GWAS and the MCS panel. Posterior-mean effect sizes were obtained from LDpred's Gibbs sampler under an infinitesimal prior, with several sparser priors estimated as sensitivity checks but not used in the final index. The shrunk weights were then applied to the harmonised MCS genotypes to produce a child-level BMI polygenic index.

Direct genetic effects (DGE) BMI PGI. SNP weights were drawn from the Tan et al. (2026) family-based BMI GWAS, which reports direct-effect estimates that have been purged of indirect parental and assortative-mating channels by conditioning on parental genotypes (Section 2). Of the 658,535 variants in the summary statistics, 585,466 with non-missing direct-effect estimates were retained. Strand-ambiguous SNPs (A/T and C/G) were excluded because allele orientation cannot be resolved unambiguously against the MCS panel, and variant coordinates were lifted over from GRCh37 to GRCh38 to align with the genotype build. The polygenic index was computed using the family-based scoring procedure of snipar (Young et al., 2022), which produces child, maternal, and paternal indices on a common scale within each family.

Estimation sample. The analysis sample of 5,446 children (Table 4) comprises children with non-missing BMI, non-missing SES, non-missing control variables, and a computable DGE PGI—that is, children for whom both parents were either directly genotyped or successfully imputed under the procedure described in Appendix 7.5. A subset of 2,707 children have both parents genotyped directly (the "full-trio" subsample used in the robustness analysis of Table 12); the remaining children have at least one imputed parent. All child, maternal, and paternal BMI PGIs are standardised to mean zero and unit standard deviation in the estimation sample.

B. Mendelian imputation of missing parental genotypes

The family-based identification strategy requires parental genotypes for two distinct purposes: as conditioning variables in the main regressions (the parental controls in Section 2), and as inputs to the family-based GWAS weights underlying the DGE PGI (Tan et al., 2026; Young et al., 2022). In the MCS, maternal genotypes are available for the great majority of genotyped children, but a substantial share of fathers were not genotyped. Two responses are possible: restrict the sample to fully observed child–mother–father trios, or recover the missing paternal genotypes using the Mendelian-transmission imputation procedure of Young et al. (2022). The main analysis adopts the second strategy and verifies that the central findings are reinforced—not artefactually generated—when the trio-only sample is used (Table 12). This appendix describes the imputation procedure, its statistical properties, and its interaction with the identification strategy.

Why imputation rather than trio-only estimation. Restricting attention to fully observed trios discards roughly half of the otherwise eligible children and, more importantly, induces selection on paternal participation. Father-genotyped families differ systematically from father-missing families on measured socioeconomic characteristics (Table A2), and likely on unmeasured ones as well. A trio-only estimator therefore answers a different question—the gradient among children whose fathers were retained in the genotyped sample—and is vulnerable to selection bias on precisely the dimension (socioeconomic status) that the paper studies. Imputation preserves the full eligible sample at the cost of additional sampling variance, while leaving the identification logic intact.

The imputed quantity. For each missing paternal genotype $g_{pi\ell}$ at SNP ℓ, the imputed value is the conditional expectation of the true paternal genotype given the available family information,

$$\hat{g}_{pi\ell} \;=\; \mathbb{E}\big[\, g_{pi\ell} \,\big|\, g_{ci\ell},\, g_{mi\ell},\, f_\ell \,\big], \tag{22}$$

where $g_{ci\ell}$ and $g_{mi\ell}$ denote the child's and the mother's allele counts and f_ℓ denotes the population minor-allele frequency. The conditional expectation in (22) is computed under the assumptions of Mendelian transmission—each parent independently transmits one of two alleles to the child with probability one half—and Hardy–Weinberg equilibrium for the untransmitted paternal allele. Given the child's and mother's genotypes, the maternally transmitted allele can in most configurations be identified, leaving the paternally transmitted allele as a known quantity and the untransmitted paternal allele as the only unobserved component; that residual allele is replaced by its population expectation f_ℓ. When both parents are missing for a child with one or more genotyped siblings, an analogous calculation conditions on the sibling configuration; children for whom no such information is available are excluded from the estimation sample.

Properties of the imputed PGI. Aggregating the imputation across SNPs by applying the same weights used to construct the standard or DGE PGI yields an imputed parental polygenic index that inherits the conditional-expectation structure of (22). By construction this index is unbiased for the true parental PGI in the sense that

$$\mathbb{E}[\,\hat{\mathrm{PGI}}_{pi} \mid g_{ci},\, g_{mi}\,] \;=\; \mathbb{E}[\,\mathrm{PGI}_{pi} \mid g_{ci},\, g_{mi}\,],$$

and it depends only on pre-determined genetic and pedigree information. In particular, the imputation does not use the BMI outcome or any socioeconomic covariate, so the imputed parental PGI is mean-independent of (y_i, S_i) given the family genetic information. This is the core reason that imputation does not threaten identification: the segregation residual that drives the within-family identification of δ_ℓ is constructed from the child's own allele count and the parental conditioning set, and replacing the conditioning set with its conditional expectation under Mendelian inheritance preserves the conditional-mean restriction $\mathbb{E}[u_{i\ell} \mid \cdot] = 0$.

Operational consequence: precision, not bias. The substantive cost of imputation is a loss of statistical precision rather than a shift in the estimand. Because $\hat{g}_{pi\ell}$ averages over the unknown untransmitted paternal allele, it is a noisier proxy for $g_{pi\ell}$ than the directly observed allele count would be. This widens standard errors on paternal coefficients and, to a lesser extent, on coefficients that interact with paternal terms; it does not introduce a systematic shift in any coefficient. The trio-only robustness analysis confirms this prediction: estimates of the standard-PGI $\times$ SES interaction obtained on the fully observed trio sample are larger in magnitude than the full-sample estimates, consistent with imputation attenuating rather than manufacturing the moderation pattern (Table 12).

Composition of the analysis sample. Among the 5,446 children in the estimation sample, 2,707 have both parents genotyped directly and 2,739 have at least one parent recovered through imputation. Table A2 reports descriptive comparisons of these two subsamples on polygenic indices, socioeconomic status, and other covariates. The imputed-parent subsample is concentrated at the lower end of the socioeconomic distribution—roughly one in five imputed-parent children fall in the bottom income quintile, against fewer than one in ten in the trio sample—which is the substantive reason for preferring the imputed-inclusive sample over a trio-only restriction. Excluding these families would deliver an estimand defined over a relatively narrow upper-SES slice of the population and would discard precisely the variation in socioeconomic status the paper studies.

C. Supplementary tables

Table A1: Child PGIs, socioeconomic status, and BMI (conventional OLS standard errors)

	Dependent variable: BMI_{it}						
	(1) PGI	(2) SES	(3) + Controls	(4) + PGI×SES	(5) + Parents	(6) + Parent×Child	(7) + Parent×SES
Panel A: Standard PGI							
Child PGI	0.921***		0.947***	0.956***	0.883***	0.889***	0.900***
	(0.025)		(0.021)	(0.021)	(0.026)	(0.026)	(0.027)
SES_c		-0.303***	-0.189***	-0.189***	-0.181***	-0.180***	-0.176***
		(0.020)	(0.016)	(0.016)	(0.016)	(0.016)	(0.016)
Mother PGI					0.148***	0.151***	0.145***
					(0.025)	(0.025)	(0.025)
Father PGI					0.013	0.011	-0.027
					(0.028)	(0.028)	(0.030)
Child PGI × SES_c				-0.042***	-0.040***	-0.028*	-0.070***
				(0.014)	(0.014)	(0.015)	(0.019)
Mother PGI × Child PGI						0.070***	0.070***
						(0.017)	(0.017)
Father PGI × Child PGI						0.016	0.028
						(0.020)	(0.020)
Mother PGI × SES_c							0.030
							(0.018)
Father PGI × SES_c							0.096***
							(0.022)
Observations	25,997	25,997	25,997	25,997	25,997	25,997	25,997
R^2	0.049	0.009	0.442	0.442	0.443	0.443	0.444
Panel B: DGE PGI							
Child DGE PGI	0.092***		0.142***	0.141***	0.096***	0.096***	0.095***
	(0.028)		(0.035)	(0.035)	(0.034)	(0.034)	(0.034)
SES_c		-0.303***	-0.289***	-0.289***	-0.210***	-0.211***	-0.210***
		(0.020)	(0.016)	(0.016)	(0.016)	(0.016)	(0.016)
Mother PGI					0.584***	0.584***	0.590***
					(0.022)	(0.022)	(0.022)
Father PGI					0.428***	0.429***	0.406***
					(0.026)	(0.026)	(0.028)
DGE PGI × SES_c				0.004	-0.005	-0.009	-0.008
				(0.016)	(0.016)	(0.016)	(0.016)
Mother PGI × DGE PGI						-0.046**	-0.048**
						(0.022)	(0.022)
Father PGI × DGE PGI						0.043	0.046*
						(0.026)	(0.026)
Mother PGI × SES_c							-0.028*
							(0.015)
Father PGI × SES_c							0.049**
							(0.020)
Observations	25,997	25,997	25,997	25,997	25,997	25,997	25,997
R^2	0.000	0.009	0.397	0.397	0.419	0.419	0.419

Notes: Appendix replication of Table 6 using conventional OLS standard errors in place of child-level clustered standard errors. Point estimates are identical to the clustered SE version; only standard errors and significance stars differ because OLS treats each child–wave observation as independent, understating uncertainty in this panel setting. The clustered SE version (Table 6) is the preferred specification. Significance: * 10%, ** 5%, *** 1%.

Table A2: Descriptive comparison of real-trio and imputed-parent subsamples within the estimation sample

	Real trio ($N = 2{,}707$)	**Has imputed** ($N = 2{,}739$)	**Diff.**	p**-value**
Polygenic indices (child-level, mean)				
Standard BMI PGI (child)	−0.085	+0.033	−0.119	< 0.001
DGE BMI PGI (child)	−0.008	+0.018	−0.027	0.33
Mother BMI PGI (standard)	−0.100	+0.000	−0.100	< 0.001
Father BMI PGI (standard)	−0.073	−0.024	−0.049	0.03
Socioeconomic status				
SES quintile (1–5, mean)	3.449	2.912	+0.537	< 0.001
Quintile 1 share (%)	8.8	20.7	−11.9	< 0.001
Quintile 5 share (%)	26.3	16.8	+9.5	< 0.001
Demographics and BMI				
Female (proportion)	0.507	0.513	−0.006	0.67
BMI mean across waves	19.27	19.65	−0.39	< 0.001
Waves per child	4.82	4.72	+0.10	< 0.001

Notes: Comparison of children in the full BMI estimation sample ($N = 5{,}446$) split into the real-trio subsample (both parents directly genotyped) and the imputed-parent subsample (Mendelian imputation of at least one parent required for inclusion). Polygenic indices are child-level z-scores standardised in the full genotyped sample before sample restrictions. SES quintile is the modified OECD equivalised household income quintile at baseline. BMI mean is averaged across the waves each child contributes. p-values from two-sample t-tests for continuous variables and χ^2 tests for proportions. The quintile 1 share and quintile 5 share rows report the percentage of each subsample falling in the lowest and highest income quintile respectively; the full five-quintile distribution differs between the two groups at $p < 0.001$ by a single χ^2 test.

References

Alemu, R., Terskaya, A., Howell, M., Guan, J., Sands, H., Kleinman, A., et al. (2025). An updated Polygenic Index Repository: Expanded phenotypes, new cohorts, and improved causal inference. *bioRxiv*. Preprint.

Amin, V., Dunn, P., and Spector, T. (2018). Does education attenuate the genetic risk of obesity? evidence from u.k. twins. *Economics and Human Biology*, 31:200–208.

Autret, K. and Bekelman, T. A. (2024). Socioeconomic status and obesity. *Journal of the Endocrine Society*, 8(11):bvae176.

Baker, S., Biroli, P., van Kippersluis, H., and von Hinke, S. (2024). Advantageous early-life environments cushion the genetic risk for ischemic heart disease. *Proceedings of the National Academy of Sciences*, 121(27):e2314056121.

Bann, D., Wright, L., Hardy, R., Williams, D. M., and Davies, N. M. (2022). Polygenic and socioeconomic risk for high body mass index: 69 years of follow-up across life. *PLOS Genetics*, 18(7):e1010233.

Barcellos, S. H., Carvalho, L. S., and Turley, P. (2018). Education can reduce health differences related to genetic risk of obesity. *Proceedings of the National Academy of Sciences of the United States of America*, 115(42):E9765–E9772.

Becker, J., Burik, C. A. P., Goldman, G., Wang, N., Jayashankar, H., Bennett, M., Belsky, D. W., Karlsson Linnér, R., Ahlskog, R., Okbay, A., et al. (2021). Resource profile and user guide of the Polygenic Index Repository. *Nature Human Behaviour*, 5:1744–1758.

Benjamin, D. J., Cesarini, D., Turley, P., and Young, A. S. (2024). Social-science genomics: Progress, challenges, and future directions. NBER Working Paper No. 32404.

Bierut, L., Biroli, P., Galama, T. J., and Thom, K. (2023). Challenges in studying the interplay of genes and environment: A study of childhood financial distress moderating genetic predisposition for peak smoking. *Journal of Economic Psychology*, 98:102636.

Biroli, P., Galama, T., von Hinke, S., van Kippersluis, H., Rietveld, C. A., and Thom, K. (2026). The economics and econometrics of gene–environment interplay. *Review of Economic Studies*, 93:144–180.

Breinholt, A. and Conley, D. (2023). Child-driven parenting: Differential early childhood investment by offspring genotype. *Social Forces*, 102(1):310–329.

Carter, A. R., Harrison, S., Gill, D., Davey Smith, G., Taylor, A. E., Howe, L. D., and Davies, N. M. (2022). Educational attainment as a modifier for the effect of polygenic scores for cardiovascular risk factors: Cross-sectional and prospective analysis of uk biobank. *International Journal of Epidemiology*, 51(3):885–897.

de Roo, M., Hartman, C., Veenstra, R., Nolte, I. M., Meier, K., Vrijen, C., and Kretschmer, T. (2023). Gene-environment interplay in the development of overweight. *Journal of Adolescent Health*, 73(3):574–581.

Dias Pereira, R., Rietveld, C. A., and van Kippersluis, H. (2025). The interplay between maternal smoking and genes in offspring birth weight. *Journal of Human Resources*, 60(2):400–433.

Fletcher, J. M. and Lu, Q. (2021). Health policy and genetic endowments: Understanding sources of response to minimum legal drinking age laws. *Health Economics*, 30(1):194–203.

Furuya, S., Fletcher, J. M., Zhao, Z., Sun, Z., and Lu, Q. (2022). Detecting genetic heterogeneities in response to trauma: The case of 9/11. *SSM - Mental Health*, 2:100044.

Houmark, M. A., Ronda, V., Agerbo, E., Mortensen, P. B., and Rosholm, M. (2025). Genetic and socioeconomic achievement gaps in elementary school. *The Economic Journal*, 135(672):2426–2454.

Howe, L. J., Nivard, M. G., Morris, T. T., Hansen, A. F., Rasheed, H., Cho, Y., Chittoor, G., Ahlskog, R., Lind, P. A., Palviainen, T., et al. (2022). Within-sibship genome-wide association analyses decrease bias in estimates of direct genetic effects. *Nature Genetics*, 54:581–592.

Hüls, A., Wright, M. N., Bogl, L. H., Kaprio, J., Lissner, L., Molnár, D., Moreno, L. A., De Henauw, S., Siani, A., Veidebaum, T., Ahrens, W., Pigeot, I., and Foraita, R. (2021). Polygenic risk for obesity and its interaction with lifestyle and sociodemographic factors in european children and adolescents. *International Journal of Obesity*, 45(6):1321–1330.

Isungset, M. A., Conley, D., Zachrisson, H. D., Ystrom, E., Havdahl, A., Njolstad, P. R., and Lyngstad, T. H. (2022). Social and genetic associations with educational performance in a Scandinavian welfare state. *Proceedings of the National Academy of Sciences*, 119(25):e2201869119.

Karlsson Linnér, R., Biroli, P., Kong, E., Meddens, S. F. W., Wedow, R., Fontana, M. A., Lebreton, M., Tino, S. P., Abdellaoui, A., Hammerschlag, A. R., Nivard, M. G., Okbay, A., Rietveld, C. A., Timshel, P. N., Trzaskowski, M., de Vlaming, R., et al. (2019). Genome-wide association analyses of risk tolerance and risky behaviors in over 1 million individuals identify hundreds of loci and shared genetic influences. *Nature Genetics*, 51(2):245–257.

Kong, A., Thorleifsson, G., Frigge, M. L., Vilhjalmsson, B. J., Young, A. I., Thorgeirsson, T. E., Benonisdottir, S., Oddsson, A., Halldorsson, B. V., Masson, G., Gudbjartsson, D. F., Helgason, A., Bjornsdottir, G., Thorsteinsdottir, U., and Stefansson, K. (2018). The nature of nurture: Effects of parental genotypes. *Science*, 359(6374):424–428.

Lindeboom, M., Lundborg, P., and van der Klaauw, B. (2010). Assessing the impact of obesity on labor market outcomes. *Economics and Human Biology*, 8(3):309–319.

Marees, A. T., Smit, D. J. A., Abdellaoui, A., Nivard, M. G., van den Brink, W., Denys, D., Galama, T. J., Verweij, K. J. H., and Derks, E. M. (2021). Genetic correlates of socio-economic status influence the pattern of shared heritability across mental health traits. *Nature Human Behaviour*, 5:1065–1073.

Martin, A. R., Kanai, M., Kamatani, Y., Okada, Y., Neale, B. M., and Daly, M. J. (2019). Clinical use of current polygenic risk scores may exacerbate health disparities. *Nature Genetics*, 51(4):584–591.

Mason, K. E., Palla, L., Pearce, N., Phelan, J., and Cummins, S. (2020). Genetic risk of obesity as a modifier of associations between neighbourhood environment and body mass index: An observational study of 335 046 uk biobank participants. *BMJ Nutrition, Prevention & Health*, 3(2):247–255.

Meddens, S. F. W., de Vlaming, R., Bowers, P., Burik, C. A. P., Karlsson Linnér, R., Lee, C., Okbay, A., Turley, P., Rietveld, C. A., Fontana, M. A., et al. (2021). Genomic analysis of diet composition finds novel loci and associations with health and lifestyle. *Molecular Psychiatry*, 26:2056–2069.

Morris, T. T., von Hinke, S., Pike, L., Ingram, N. R., Davey Smith, G., Munafo, M. R., and Davies, N. M. (2024). Implications of the genomic revolution for education research and policy. *British Educational Research Journal*, 50(3):923–943.

Mostafavi, H., Harpak, A., Agarwal, I., Conley, D., Pritchard, J. K., and Przeworski, M. (2020). Variable prediction accuracy of polygenic scores within an ancestry group. *eLife*, 9:e48376.

Muslimova, D., van Kippersluis, H., Rietveld, C. A., von Hinke, S., and Meddens, S. F. W. (2024). Gene-environment complementarity in educational attainment. *Journal of Labor Economics*. Forthcoming.

Okbay, A., Baselmans, B. M. L., De Neve, J.-E., Turley, P., Nivard, M. G., Fontana, M. A., Meddens, S. F. W., Karlsson Linnér, R., Rietveld, C. A., Derringer, J., Gratten, J., Lee, J. J., Liu, J. Z., de Vlaming, R., et al. (2016). Genetic variants associated with subjective well-being, depressive symptoms, and neuroticism identified through genome-wide analyses. *Nature Genetics*, 48(6):624–633.

Pereira, R. D., Biroli, P., Galama, T., von Hinke, S., van Kippersluis, H., Rietveld, C. A., and Thom, K. (2022). Gene-environment interplay in the social sciences. In *Oxford Research Encyclopedia of Economics and Finance*. Oxford University Press.

Pereira, R. D. and van Kippersluis, H. (2025). Intergenerational education persistence: Evidence from molecular genetic data. Tinbergen Institute Discussion Paper No. TI 2025-057/V.

Privé, F., Arbel, J., and Vilhjálmsson, B. J. (2021). Ldpred2: Better, faster, stronger. *Bioinformatics*, 36(22-23):5424–5431.

Segal, A. B., Huerta, M. C., Aurino, E., and Sassi, F. (2021). The impact of childhood obesity on human capital in high-income countries: A systematic review. *Obesity Reviews*, 22(1):e13104.

Shi, J., Park, J.-H., Duan, J., Berndt, S. T., Moy, W., Yu, K., Song, L., Wheeler, W., Hua, X., Silverman, D., Garcia-Closas, M., Hsiung, C. A., Figueroa, J. D., Cortessis, V. K., Malats, N., Karagas, M. R., Vineis, P., Chang, I.-S., Lin, D., and Chatterjee, N. (2016). Winner's curse correction and variable thresholding improve performance of polygenic risk modeling based on genome-wide association study summary-level data. *PLoS Genetics*, 12(12):e1006493.

Tan, T., Jayashankar, H., Guan, J., Nehzati, S. M., Mir, M., Bennett, M., et al. (2026). Family-GWAS reveals effects of environment and mating on genetic associations. *medRxiv*. Preprint, posted 22 January 2026.

Uchikoshi, F. and Conley, D. (2021). Gene-environment interactions and school tracking during secondary education: Evidence from the U.S. *Research in Social Stratification and Mobility*, 76:100628.

van Alten, S., Barcellos, S. H., Carvalho, L., Galama, T. J., and Aguiar Palma, M. (2025). A chip off the old block? genetics and the intergenerational transmission of socioeconomic status. NBER Working Paper No. 34208.

van den Berg, G. J., von Hinke, S., and Wang, R. A. H. (2025). Prenatal sugar exposure shapes late-life human capital and health. *PNAS Nexus*, 4(10):pgaf301.

Vilhjálmsson, B. J., Yang, J., Finucane, H. K., Gusev, A., Lindström, S., Ripke, S., Genovese, G., Loh, P.-R., Bhatia, G., Do, R., Hayeck, T., Won, H.-H., Kathiresan, S., Pato, M., Pato, C., Tamimi, R., Stahl, E., Zaitlen, N., Pasaniuc, B., Belbin, G., Kenny, E. E., Schierup, M. H., De Jager, P., Patsopoulos, N. A., McCarroll, S., Daly, M., Purcell, S., Chasman, D., Neale, B., Goddard, M., Visscher, P. M., Kraft, P., Patterson, N., and Price, A. L. (2015). Modeling linkage disequilibrium increases accuracy of polygenic risk scores. *American Journal of Human Genetics*, 97(4):576–592.

Wang, Y., Guo, J., Ni, G., Yang, J., Visscher, P. M., and Yengo, L. (2020). Theoretical and empirical quantification of the accuracy of polygenic scores in ancestry divergent populations. *Nature Communications*, 11:3865.

Wertz, J., Belsky, J., Moffitt, T. E., Belsky, D. W., Harrington, H., Avinun, R., Poulton, R., Ramrakha, S., and Caspi, A. (2019). Genetics of nurture: A test of the hypothesis that parents' genetics predict their observed caregiving. *Developmental Psychology*, 55(7):1461–1472.

Yengo, L., Sidorenko, J., Kemper, K. E., Zheng, Z., Wood, A. R., Weedon, M. N., Frayling, T. M., Hirschhorn, J., Yang, J., Visscher, P. M., and GIANT Consortium (2018). Meta-analysis of genome-wide association studies for height and body mass index in ∼700,000 individuals of European ancestry. *Human Molecular Genetics*, 27(20):3641–3649.

Young, A. I., Nehzati, S. M., Benonisdottir, S., Okbay, A., Jayashankar, H., Lee, C., Cesarini, D., Benjamin, D. J., Turley, P., and Kong, A. (2022). Mendelian imputation of parental genotypes improves estimates of direct genetic effects. *Nature Genetics*, 54(6):897–905.